

Unequal Access to Equalizing Opportunities*

Variation in the Costs, Accessibility, and Benefits of English-Learning Opportunities in India

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Abstract

English learning is often viewed as a potential tool for economic mobility in India, where English skills command substantial wage premia (Azam et al. 2013; Refeque and Azad 2022). I study whether the conditions surrounding households’ investments in children’s English skills are themselves related to inequality. I organize the analysis around three conditions: inherited linguistic conditions, access to local English-learning opportunities, and the economic setting in which English skills may be rewarded. Using harmonized district data spanning languages, schooling, consumption, migration, household human capital, and sectoral employment, I find inequality across all three conditions. Linguistic conditions are persistent and associated with language-learning behavior; formal English-medium schooling is strongly organized by state institutions; and access to English-medium and private schooling remains socially stratified within districts. English-intensive schooling is associated with later local prosperity, particularly where complementary human capital and urban capacity are greater, while inherited linguistic conditions also accompany broader skill-intensive local development. Linguistic distance does not provide persuasive within-state variation in English-medium schooling, so I do not interpret the schooling associations as causal effects of English-medium instruction. A high-return educational opportunity need not equalize opportunity when access to it and the resources required to benefit from it are themselves unequal.

1 Introduction

The history of economics highlights *education* as one rich answer to the question of how households can improve their children’s future prospects (Mincer 1974; Card 1999; Björklund and Kjellström 2002; Bhandari and Bordoloi 2006). The history of India highlights another: *English*. In its capacity as a global lingua franca (Melitz 2016), English is perceived as expanding mobility (Itani et al. 2015) and opening pathways to education at prestigious universities and employment at multinational firms (Ramanathan 2014; Bedi 2019). English also serves as a

*Acknowledgments: I thank the professors who have provided feedback on this project and the Internet Archive for preserving source materials that would otherwise have been inaccessible. Large language models assisted with research and editing; I remain responsible for the analysis and writing.

domestic lingua franca in a country with extraordinary linguistic diversity (Clingingsmith 2014). The expansion of information technology and internationally traded services increased the demand for workers able to use English (Mankiw and Swagel 2006; Shastry 2012).

Empirical study gives households a concrete reason to care. Using nationally representative household data, Azam et al. (2013) find that English fluency was associated with a 34% hourly wage premium for men and a 22% premium for women in 2005; even limited English proficiency was associated with premia of 13% and 10%, respectively. Refeque and Azad (2022) later document positive English earnings differentials through 2011–12. Using West Bengal’s 1983–2004 withdrawal of English teaching from public primary schools, Chakraborty and Bakshi (2016) estimate that cohorts more exposed to the policy experienced lower subsequent wages. These findings help explain the demand for *English-medium instruction (EMI)*, the teaching of school subjects in English (Gooptu and Mukherjee 2023; Lahoti and Mukhopadhyay 2019; Chattopadhyay and Roy 2017). EMI has consequently been promoted as a tool of social development (National Knowledge Commission 2009) and described by advocates as a route to mobility from caste-linked occupational constraints (Mathur 2013; Mathew and Srivastava 2011; Vij 2020), against a background in which caste remains associated with wages, employment, and occupation (Madheswaran and Singhari 2016; Munshi and Rosenzweig 2006).

Less corroborated, however, are the conditions surrounding a household’s decision to invest in a child’s English skills. I ask whether those conditions are associated with the same inequalities that English learning is often expected to help households escape. I study inequalities by social group, religion, parental education, household resources, and urban or rural residence. If access to English learning is systematically related to these characteristics, then a high economic payoff to English skills need not make English learning equalizing.

I use the household investment problem as an organizing lens for three potential determinants of English-learning investment. I do not estimate a structural household-choice model; instead, I study three empirically observable conditions surrounding the decision. The first is **inherited linguistic conditions**: historically persistent features of the languages spoken locally that can affect the relative cost of learning English rather than another bridge language, meaning a language used for communication across language groups. I study this condition using several measures of local linguistic distance from Hindi; Shastry (2012) provides the economic motivation for treating distance from Hindi as a measure of the relative cost of acquiring English rather than Hindi as a bridge language. The second is **current access to English-learning opportunities**: the local schooling options households can choose from and their realized access to those options, including enrollment, English-medium schooling, and public versus private provision. The third is **anticipated economic returns to English learning**. I do not observe household beliefs directly. I use established evidence on the labor-market value of English to characterize the returns households could observe when forming expectations. I then use later district outcomes to ask whether

English-intensive schooling is associated with subsequent local welfare and whether that association differs with local conditions. As an organizing assumption, I treat expectations as adaptive: households form beliefs from economic conditions and returns they can observe rather than from a model of all future states.

The first of these measures is deliberately narrower than a child’s complete language environment. Rowe and Weisleder (2020) describe language development as occurring within nested contexts that range from societal norms and policies to the day-to-day interactions through which children encounter language. District linguistic composition cannot capture English or Hindi exposure through film, television, social media, migration, peers, or household interactions. I therefore use *inherited linguistic conditions* for the historically rooted district measures studied here, rather than treating them as a comprehensive measure of children’s language exposure.

All three conditions are studied locally. The district is the most disaggregated geography at which I can harmonize the longitudinal sources used throughout the paper. This level also corresponds to the local schooling markets and economic outcomes I observe. Earlier work reported limited interdistrict mobility in India: Topalova (2005) documents low decadal interdistrict migration, especially in rural areas and for employment-related moves, while Chakraborty and Bakshi (2016) used similarly low mobility in parts of India to relate schooling location to later residence. I do not assume that mobility is negligible. More recent work emphasizes the roles of education and internal migration in responses to India’s information-technology expansion (Ghose 2026), and my own results associate inherited linguistic conditions with the skill composition of recent work migrants; I return to those results in Section B. District outcomes should therefore be interpreted as outcomes for places and their changing populations, rather than as individual returns for the children observed in an earlier survey.

The results show that the conditions surrounding English-skill investment are themselves unequally distributed. The relative cost of acquiring a bridge language is partly inherited, and language behavior varies with these inherited linguistic conditions. Formal English-medium schooling is strongly organized by state boundaries, particularly in the public sector. Within the same district, social-group differences remain large in English-medium and private-school participation, with larger differences in urban schooling markets. English-intensive schooling is more strongly associated with later local consumption than enrollment alone, and this association is stronger in districts with higher initial human capital and urbanization. The inherited linguistic gradient is also associated with later changes in human capital, the skill composition of recent work migrants, financial access, and sectoral composition. These broader associations are selective rather than evidence of general employment expansion.

Taken together, the evidence limits the equalizing promise of a high-return educational opportunity. A child may face high future payoffs to English skills while the household inherits a less favorable linguistic starting point, faces a more restricted effective schooling choice set, or lives in a district with fewer complementary resources through which English skills can be converted into economic opportunity. English skills can be valuable without

access to English learning being equal. A high-return educational opportunity does not automatically equalize opportunity when entry into it, and the resources needed to benefit from it, are themselves unequally distributed.

2 Background

It is by combining education and English that the demand for *English-medium instruction (EMI)* becomes intelligible. Language skills are human capital ([Chiswick and Miller 1995](#); [Azam et al. 2013](#)), and households may reasonably value schooling that they expect to build a skill rewarded in education and the labor market. Yet learning English and learning *through* English are different investments. EMI uses English to teach school subjects; Hindi-medium and mother-tongue-medium schools use other languages to teach those subjects. The economic value of English proficiency therefore does not, by itself, establish the educational value of using English as the medium of instruction.

The original paper emphasized this distinction because instruction in a familiar language can improve learning. In India, Telugu-medium primary-school students performed better in mathematics than otherwise comparable English-medium students in longitudinal data from Andhra Pradesh and Telangana ([Nair 2015](#)), while work on multilingual education documents the difficulties created when children enter school through a language they do not use at home ([Mohanty 2022](#)). Historical evidence reaches a similar conclusion from a different setting. Before India’s linguistic state reorganization, districts whose local language differed from the official language had substantially lower literacy and college graduation; those differences narrowed after state borders and official languages were reorganized ([Jain 2017](#)). These results do not imply that English instruction is uniformly harmful. They establish the narrower point needed here: a valuable language skill and the language used to teach non-language subjects are not the same educational input.

Reported school medium is also an imperfect measure of the language children hear in class. Lahoti and Mukhopadhyay ([2019](#)) find that more than half of children described as attending English-medium schools in their rural sample were actually being taught in the dominant regional language; some other schools used English books while teachers explained them in the regional language. Similar concerns appear in qualitative accounts of English-medium schooling ([Mathew and Srivastava 2011](#); [Bhattacharya 2013](#)). I therefore interpret reported EMI as exposure to an English-medium schooling category, not as a direct measure of how much English a child hears, speaks, or learns.

A second distinction concerns who manages the school. The demand for both higher-quality education in general and EMI in particular has, since at least the 2000s, increasingly been met by private schools ([James and Woodhead 2014](#); [Srivastava 2013](#); [Ramanathan 2016](#); [Chattopadhyay and Roy 2017](#)). But demand for private schooling should not be equated with demand for EMI. In the 2014 survey summarized by Joshi and Kumar ([2017](#)), households

choosing private schools much more often cited a better learning environment or better education than government schools than English as the medium of instruction. The two choices can coincide without representing the same household preference.

Private schooling also changes the resource constraint facing a household. Even schools described as low-fee can absorb a substantial share of poor households' income (Endow 2019), and private enrollment is strongly selected on household characteristics (Muralidharan and Kremer 2008; Muralidharan and Sundararaman 2015; Raju 2023). That selection makes raw public-private comparisons difficult to interpret as differences in school effectiveness. In a randomized voucher experiment in Andhra Pradesh, Muralidharan and Sundararaman (2015) find no difference between lottery winners and losers in Telugu, mathematics, English, or science and social studies after two and four years, despite large cross-sectional public-private differences; private schools did, however, teach Hindi and operated at substantially lower cost. School management is therefore not a sufficient measure of school quality.

Nor is school choice independent of social institutions. In Mumbai, Munshi and Rosenzweig (2006) show that caste networks continued to direct lower-caste boys toward local-language schools associated with traditional occupations even as returns to nontraditional white-collar work rose, whereas lower-caste girls shifted much more rapidly toward English schools. This evidence is especially relevant here because it links the language of schooling to social position without treating English medium or private management as interchangeable measures of advantage.

The empirical analysis therefore keeps the two dimensions separate. Public versus private schooling describes school management and the household's sector choice; English versus non-English medium describes the language through which subjects are taught. A private school need not provide effective instruction in English, and an English-medium label need not indicate higher school quality. I consequently study enrollment, EMI conditional on enrollment, all-child EMI exposure, public EMI, and private EMI as distinct measures rather than collapsing them into a single index of educational opportunity.

3 Data and Variables

The data work is a substantial part of this paper. I place survey, Census, school-report, and economic data observed under changing district boundaries onto a common Census-2001 geography; recover district-level DISE schooling reports from archived spreadsheets and PDFs; construct several district measures of linguistic distance from the Census language tables; and make consumption comparable across older NSS rounds and the newer Household Consumption Expenditure Surveys (HCES) by reconciling survey definitions, district identities, and price indices. The construction details and source checks used by the final paper are integrated into Section A; identification checks are reported in Section D.

Geography and source coverage. District boundaries change often enough that a district name cannot be treated as a stable longitudinal identifier. I use Census-2001 districts as the reference geography and reconcile later and earlier source units using reviewed district-lineage evidence. The lineage system is informed by Census materials and independent district-harmonization resources, including the India State and District Evolution Database, the Indian Census Data Collection, the India District Changes Tracker, and SHRUG (Jolad et al. 2026; Jolad and Singh 2026; Jaacks et al. 2020; Asher et al. 2021). It supports 593 Census-2001 reference districts; the principal schooling panel contains 573 because the 2007–08 education survey does not yield the required measures for every reference district. The main analysis therefore uses district aggregates rather than treating survey respondents as a longitudinal panel of individuals.

The principal schooling measures come from the 2007–08 National Sample Survey (NSS) 64th Round, *Participation and Expenditure in Education* (National Sample Survey Office 2011). I use survey weights when forming district aggregates. DISE provides an independent school-reported source on medium of instruction, management, enrollment, and school characteristics. I reconstruct the DISE district series from report cards covering 2005–06 through 2007–08 that were preserved online after the original publication pages disappeared (Mehta 2022; District Information System for Education 2017). Census-2001 language tables provide mother-tongue populations and language behavior (Office of the Registrar General and Census Commissioner of India 2001). Later sections also use Census-2001 and Census-2011 migration, household, housing, and worker tables, the 2005 and 2013–14 Economic Censuses (Central Statistics Office 2008, 2016), the 2009–10 NSS employment survey (National Sample Survey Office 2010), and the 2017–18 Periodic Labour Force Survey (National Statistical Office 2019).

Current access to English-learning opportunities. The NSS schooling variables separate participation in school from medium of instruction. Let N_d be the survey-weighted number of children ages 5–19 in district d , E_d the number enrolled, K_d the number enrolled for whom medium of instruction is observed, and M_d the number in K_d enrolled in English-medium instruction (EMI). I call

$$\text{Enrollment}_d = 100 \frac{E_d}{N_d} \tag{1}$$

the **enrollment rate**. This is the extensive margin: whether a child is enrolled at all. Among enrolled children whose medium of instruction is observed, I call

$$\text{EnglishMediumShare}_d = 100 \frac{M_d}{K_d} \tag{2}$$

the **English-medium share among enrolled children**. This is the conditional medium-of-instruction margin

among children with a reported medium. Finally,

$$\text{EnglishMediumExposure}_d = 100 \frac{M_d}{N_d} \quad (3)$$

is **English-medium exposure among all children**. It combines enrollment and medium choice and is therefore lower than the conditional English-medium share whenever some children are out of school. I also distinguish **public-school English-medium exposure** and **private-school English-medium exposure**, each defined as the share of all children enrolled in English-medium schooling in the relevant management sector, and **private-school enrollment**, the share of enrolled children attending private schools. These names refer to realized schooling choices rather than a complete menu of learning options: the NSS does not measure private tutoring or other supplemental English instruction.

Inherited linguistic conditions. The primary language measure begins with Shastri’s 0–5 ordinal scale of separation from Hindi, developed with linguist Jay Jasanoff (Shastri 2012). Shastri uses the scale to motivate distance from Hindi as a measure of the relative cost of acquiring Hindi rather than English as a bridge language. For each mapped, non-English mother tongue l in district d , let S_{ld} denote its Census-2001 speaker population and D_l its Shastri distance from Hindi. The **speaker-weighted distance from Hindi** is

$$\bar{D}_d = \frac{\sum_l S_{ld} D_l \mathbf{1}\{D_l > 0\}}{\sum_l S_{ld} \mathbf{1}\{D_l > 0\}}. \quad (4)$$

The positive-distance restriction prevents the size of the Hindi-speaking population from mechanically determining the mean distance among languages that require movement away from Hindi. I use **linguistic distance from Hindi** as a shorter name for this speaker-weighted measure in the main text. A second measure, the **share speaking languages distant from Hindi**, is the percentage of all mother-tongue speakers in the district whose mapped language has Shastri distance at least three. Unlike the mean-distance measure, this statistic retains the size of the distant-language population as part of the construct.

I use several alternatives when testing whether results depend on one linguistic classification. The **top-three-language distance from Hindi** is the speaker-weighted Shastri distance among the three largest mapped mother tongues in a district, corresponding to the older construction used in the original analysis. **Genealogical distance from Hindi** counts separation along the Glottolog language-family tree (Hammarström et al. 2026). **Lexical noncognacy with Hindi** is one hundred minus the percentage of cognate items shared with Hindi in the Dyen lexicostatistical data (Dyen et al. 1992; “Comparative Indo-European Database” 1997). I also reconstruct the speaker-weighted and distant-language measures using the 1991 Language Atlas for historical validation (Office

of the Registrar General and Census Commissioner, India 2004). The crosswalk and adjudication rules for Census mother tongues are documented in Section A.

Economic outcomes and predetermined controls. I use household consumption as the principal measure of local material living standards. Monthly per-capita consumption expenditure (MPCE) is household consumption expenditure divided by household size; after deflation I refer to the district mean simply as **real consumption per person**. The baseline comes from the 2004–05 NSS 61st Round (National Sample Survey Office 2007). The 2009–10 and 2011–12 NSS rounds provide intermediate endpoints (National Sample Survey Office 2012b, 2014), and HCES 2022–23 and 2023–24 provide the long-run endpoints (National Sample Survey Office 2024, 2025a). I harmonize the older and newer surveys to 2011–12 rupees using official price series, including the Reserve Bank of India and Labour Bureau series (Reserve Bank of India 2026; Labour Bureau, Government of India 2026a). Because HCES uses a three-visit questionnaire design, its household consumption measure is reconstructed from the survey’s three questionnaire components before district aggregation; Section A gives the details.

The preferred district regressions compare districts within states and condition on characteristics measured before the 2007–08 schooling outcomes. The Census-2001 controls are log population, urban population share, secondary-or-higher educational attainment among people age seven and older, Scheduled Caste share, Scheduled Tribe share, Muslim share, the agricultural share of workers, the dependency ratio, household electricity access, and log population density (Office of the Registrar General and Census Commissioner of India 2001; Asher et al. 2021). I refer to the education and urbanization measures separately when they are used as predetermined complements to English-medium schooling rather than as baseline controls.

Table~1 reports the scale and dispersion of the principal variables. Enrollment averages about 74 percent of children ages 5–19, while English-medium exposure among all children averages about 15 percent. Private-school English-medium exposure averages about 9 percent and public-school English-medium exposure about 6 percent. The speaker-weighted distance from Hindi averages 3.05 Shastry degrees across districts, and the district mean of the share speaking languages at distance three or greater is 42 percent. Real consumption per person rises from an average of about Rs. 928 per month in 2004–05 to Rs. 2,152 in 2022–23 and Rs. 2,242 in 2023–24, all expressed in 2011–12 prices.

Table 1: Core Variables and Summary Statistics. Sources: NSS 64th Round; Census of India 2001; NSS 61st Round; HCES 2022-23 and 2023-24.

Variable	Mean	SD	N	p10	p90	Year / unit
Panel A. Schooling, 2007-08:						
Enrollment	74.44	9.72	573	62.07	87.12	2007-08; % of children age 5-19
English-medium share among enrolled	18.99	25.34	573	0.83	52.98	2007-08; % of enrolled children
English-medium exposure among all children	14.88	20.33	573	0.53	43.89	2007-08; % of children age 5-19
Public-school English-medium exposure	5.55	14.26	573	0.00	15.56	2007-08; % of children age 5-19
Private-school English-medium exposure	9.20	11.58	573	0.32	25.41	2007-08; % of children age 5-19
Private-school enrollment	27.11	19.40	573	3.23	55.43	2007-08; % of enrolled children
Panel B. Inherited linguistic conditions:						
Speaker-weighted distance from Hindi	3.05	1.36	573	1.04	4.97	2001; Shastry degrees
Share speaking languages distant from Hindi	42.23	42.80	573	0.13	98.44	2001; % of mother-tongue speakers
Panel C. Predetermined district capacity:						
Secondary-plus share, age 7+	17.09	7.63	573	8.48	27.84	2001; %
Urban population share	23.27	19.11	573	6.18	47.54	2001; %
Scheduled Tribe share	16.16	25.85	573	0.00	56.59	2001; %
Panel D. Later welfare:						
Real mean consumption per person	928.07	377.75	557	464.61	1419.84	2004-05; 2011-12-price Rs/person/month
Real mean consumption per person	2152.01	618.67	550	1490.97	2952.44	2022-23; 2011-12-price Rs/person/month
Real mean consumption per person	2242.07	601.13	551	1622.74	2991.47	2023-24; 2011-12-price Rs/person/month

Note: District-level descriptive statistics; N is the number of districts with finite values. Schooling measures use NSS 2007-08 children age 5-19; linguistic distance and predetermined capacity use Census-2001 geography. Modern consumption rows summarize districts meeting the survey-design, price, and district-lineage requirements used for the preferred estimates. p10 and p90 are the 10th and 90th percentiles. Sources: NSS 64th Round; Census of India 2001; NSS 61st Round; HCES 2022-23 and 2023-24. Construction details are documented in the data appendix.

4 Inherited Linguistic Conditions

4.1 Proxies and Persistence of Linguistic Conditions

Why measure distance from Hindi rather than distance from English? India’s linguistic diversity gives languages of wider communication an important role, and both Hindi and English have served that role across groups (Clingingsmith 2014; Shastry 2012). Shastry’s argument is comparative: linguistic diversity often compels people to learn either English or Hindi, so a mother tongue that is more distant from Hindi raises the relative cost of learning Hindi and thereby lowers the opportunity cost of learning English (Shastry 2012). In the original paper’s terms, as the distance between one’s mother tongue and Hindi increases, I expect the opportunity cost of learning English over Hindi to decrease. The speaker-weighted measure defined in Section 3 therefore summarizes a historically rooted difference in the relative cost of acquiring Hindi rather than a direct measure of English proficiency.

I call these conditions *inherited* because the measured district pattern changes little across the decade for which I can reconstruct it independently. Among the 91 Census-2001 districts meeting the historical reconstruction requirements in both years, the correlation between the 1991 and 2001 speaker-weighted distance measures is 0.898; weighting districts by their 1991 populations raises the correlation to 0.928. The share of speakers whose mother tongues are at least three Shastry degrees from Hindi is similarly persistent, with a population-weighted correlation of 0.919. Figure 1 shows the district-level comparison. This persistence is consistent with Shastry’s finding that district linguistic distance is highly correlated across earlier Census years (Shastry 2012).

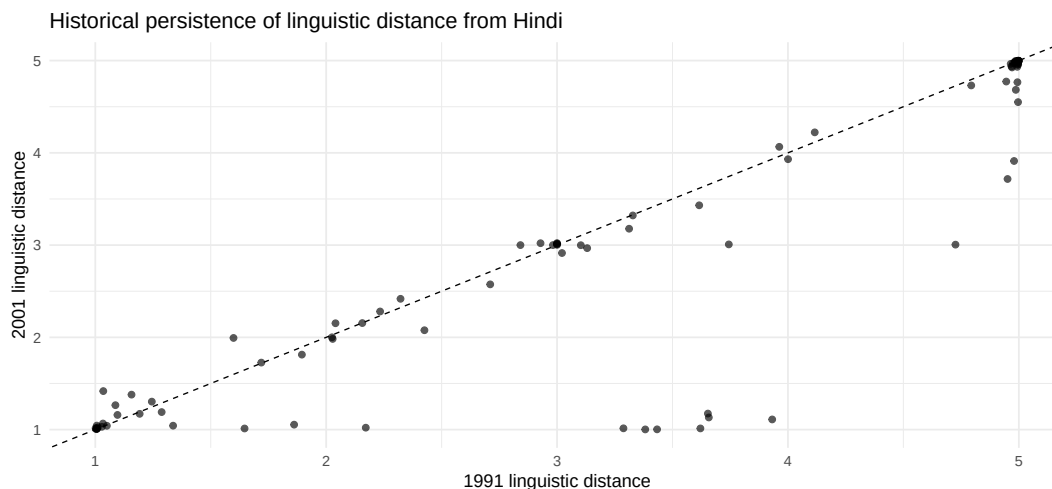


Figure 1: Persistence of speaker-weighted linguistic distance from Hindi from 1991 to 2001. Each point is a Census-2001 district meeting the historical language-reconstruction requirements in both years; the dashed line is the 45-degree line. Sources: Census of India 1991 Language Atlas ([Office of the Registrar General and Census Commissioner, India 2004](#)) and Census of India 2001 language tables ([Office of the Registrar General and Census Commissioner of India 2001](#)).

The historical reconstruction is also checked against an external source. Helms and Lim’s replication data include a Shastry-derived 1991 district linguistic-distance measure ([Helms and Lim 2025](#)). On the 148 districts where their extract and my 1991 reconstruction overlap, the two measures have a correlation of 0.989. I use this comparison as a source-reconstruction check: it verifies that two independently assembled 1991 series recover nearly the same district ranking, rather than introducing another definition of linguistic distance.

Persistence does not mean that a district’s language environment is fully summarized by these measures. Children encounter language through family members, peers, schools, media, and other social contexts; those contexts can affect both the amount and kind of language input they receive ([Rowe and Weisleder 2020](#)). Linguistic distance from Hindi therefore captures one persistent component of the relative cost of acquiring a bridge language, not the complete language environment a child inherits.

4.2 Language-Learning Behavior as a Function of Linguistic Conditions

A relative-cost measure is economically useful only if it corresponds to differences in language-learning behavior. The Census-2001 bilingualism and trilingualism tables record native-language populations and their reported subsidiary languages, allowing me to compare English acquisition, Hindi acquisition, and multilingualism across language groups ([Office of the Registrar General and Census Commissioner of India 2001](#)). The regressions in Table~2 are weighted by native-language population and compare language groups within states while conditioning

on their state population share and whether they are the state's modal language.

Table 2: Linguistic Distance and Language-Learning Behavior. Source:
Census of India 2001 bilingualism and trilingualism tables.

	All states				Hindi-belt states		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Linguistic distance from Hindi	2.234		-9.694***	-3.581***	-0.037		-9.024*
	(1.448)		(2.573)	(0.916)	(2.942)		(5.321)
Distant-language indicator		16.727***				10.165	
		(4.394)				(10.807)	
Outcome	English acquisition	English acquisition	Hindi acquisition	Multilingualism	English acquisition	English acquisition	Hindi acquisition
Population	Multilingual speakers	Multilingual speakers	Multilingual speakers	Native speakers	Multilingual speakers	Multilingual speakers	Multilingual speakers
Observations	1,566	1,566	1,566	1,587	560	560	560
Partial R^2	0.025	0.128	0.094	0.139	0.000	0.046	0.121
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language-share controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Each column is a weighted state-by-language regression. Standard errors in parentheses use HC1 heteroskedasticity-robust inference. All specifications include state fixed effects, native-language state share, an indicator for the state's most common native language, and a Hindi/Urdu reference indicator where required by the registered specification. Continuous-distance coefficients are percentage-point changes per one Shastry distance degree; the distant-language indicator equals one for languages at least three degrees from Hindi. Partial R-squared is the one-degree-of-freedom model-based partial R-squared for the reported linguistic-distance coefficient. Columns (5)-(7) apply the predeclared Hindi-belt sample restriction. These are descriptive associations with language-learning behavior rather than instrumental-variable first stages. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ Source: Census of India 2001 bilingualism and trilingualism tables.

The continuous English relationship is positive but imprecise. One additional Shastry distance degree is associated with about 2.23 percentage points more English acquisition among multilingual speakers, with $p = 0.123$. The nonlinear comparison is much stronger: speakers of languages at least three degrees from Hindi have an English-acquisition rate about 16.73 percentage points higher than speakers of the remaining classified mother tongues. This contrast is consistent with Shastry’s emphasis on nonlinear differences in the relative cost of learning English (Shastry 2012).

Hindi acquisition changes in the opposite direction. A one-degree increase in Shastry distance is associated with about 9.69 percentage points less Hindi acquisition among multilingual speakers. Yet the evidence is not a simple reallocation of a fixed group of bilingual speakers from Hindi toward English. Greater distance is also associated with a 3.58 percentage-point decline in the share of native speakers reported as multilingual. Linguistic distance is therefore associated both with which bridge language multilingual people acquire and with whether people report an additional language at all.

The relationship also depends on how linguistic difference is measured. The English-acquisition association is much weaker for the genealogical-distance and lexical-noncognacy alternatives defined in Section 3: neither produces a comparably precise continuous English relationship in the corresponding specifications. I therefore treat the strong distant-language contrast as evidence about the Shastry relative-cost construction, not as a general claim that every measure of linguistic difference predicts English acquisition.

4.3 Language-Learning Behavior as a Function of Social and Institutional Factors

The national relationship weakens considerably when I restrict attention to the Hindi-belt states. In that sample, the continuous English coefficient is essentially zero (-0.04 percentage points, $p = 0.990$), while the distant-language English contrast is about 10.16 percentage points but imprecisely estimated ($p = 0.347$). The Hindi coefficient remains negative, at about 9.02 percentage points per distance degree, but is also less precise ($p = 0.091$). Thus the national English-acquisition pattern is not reproduced within one broadly Hindi-oriented group of states.

Historical evidence for Scheduled Tribe language groups points in the same direction. In the 1991 language data (Office of the Registrar General and Census Commissioner, India 2004), greater distance is associated with more Hindi acquisition rather than more English acquisition; this pattern is stronger in the Hindi-belt restriction, although that historical comparison contains only four state clusters. I therefore use it as descriptive evidence that the destination language depends on the surrounding institutional and economic context, not as a separate causal result. This interpretation is consistent with evidence that economic incentives can induce second-language acquisition in India (Clingsmith 2014).

There is little indication that the national pattern is specific to one sex. In the Census-2001 language data (Office

of the Registrar General and Census Commissioner of India 2001), the English coefficients estimated separately for men and women are similar in magnitude, about 2.33 and 2.01 percentage points per Shastry distance degree. The more consequential variation is geographic: the same distance measure corresponds to different English-versus-Hindi behavior across institutional settings.

Taken together, the results support a narrower interpretation than a mechanical English/Hindi substitution story. Linguistic distance from Hindi captures a persistent difference in the relative cost of acquiring a widely used bridge language. It does not determine which bridge language people actually acquire. Local institutions and economic conditions help determine whether that relative-cost difference is expressed as English learning, Hindi learning, or no additional reported language. The next section asks how those local conditions appear in formal schooling.

5 Present Access to English-Learning Opportunities

5.1 Proxying for the Menu of English-Learning Opportunities

Inherited linguistic conditions can change the relative cost of acquiring English, but households can invest only through the options they can actually reach. I therefore use the schooling measures defined in Section 3 as proxies for the local menu of English-learning opportunities. The NSS 64th Round records realized household choices in 2007–08: whether children ages 5–19 are enrolled, the medium of instruction among enrolled children with a reported medium, and school management among enrolled children with reported management (National Sample Survey Office 2011). These are equilibrium outcomes of what schools are available, household preferences, and household resources; they are not direct measures of school supply.

The three main schooling margins separate different parts of that choice. The enrollment rate is the extensive margin: whether an age-eligible child is enrolled at all. The English-medium share among enrolled children is an intensive margin among children already in school and with observed medium of instruction. English-medium exposure among all children combines both, because a child must first be enrolled and then be observed in English-medium instruction. I also separate public- and private-school English-medium exposure and private-school enrollment. This distinction matters because the demand for private schooling is not synonymous with the demand for English. In the private-school literature, households often report perceived learning quality or school environment as more important motives than English medium itself (Joshi and Kumar 2017), while private-school access is strongly selected by household resources (Endow 2019; Raju 2023).

I focus on English-medium instruction because it is an observable way in which schools package English with formal education, and because demand for EMI has long been tied to aspirations for English skills (Lahoti

and Mukhopadhyay 2019). Learning English and learning *through* English are nevertheless different activities. Nominal English-medium status need not imply that classroom instruction actually occurs in English: Lahoti and Mukhopadhyay (2019) report substantial disagreement between stated and observed medium in rural private schools. My NSS measures should therefore be read as realized exposure to reported EMI, not as measures of English proficiency or instructional quality. The NSS also does not observe private tutoring or other supplemental English instruction, so the schooling measures capture only part of the English-learning menu.

DISE provides a complementary school-reported perspective. Its district aggregates describe the composition of the local school system, including enrollment by medium and management, rather than household survey choices. Because enrollment shares still combine supply with family demand, I treat DISE as closer to the supply side rather than as a pure supply measure. The agreement between NSS and DISE later in this section is useful precisely because the two sources reach the same schooling margin differently (Mehta 2022; District Information System for Education 2017).

Figure 2 shows the geography of linguistic distance and the three English-medium exposure measures. Public-school EMI is concentrated especially in northern and northeastern districts, while private-school EMI is more geographically dispersed. The descriptive contrast motivates the institutional comparisons below.

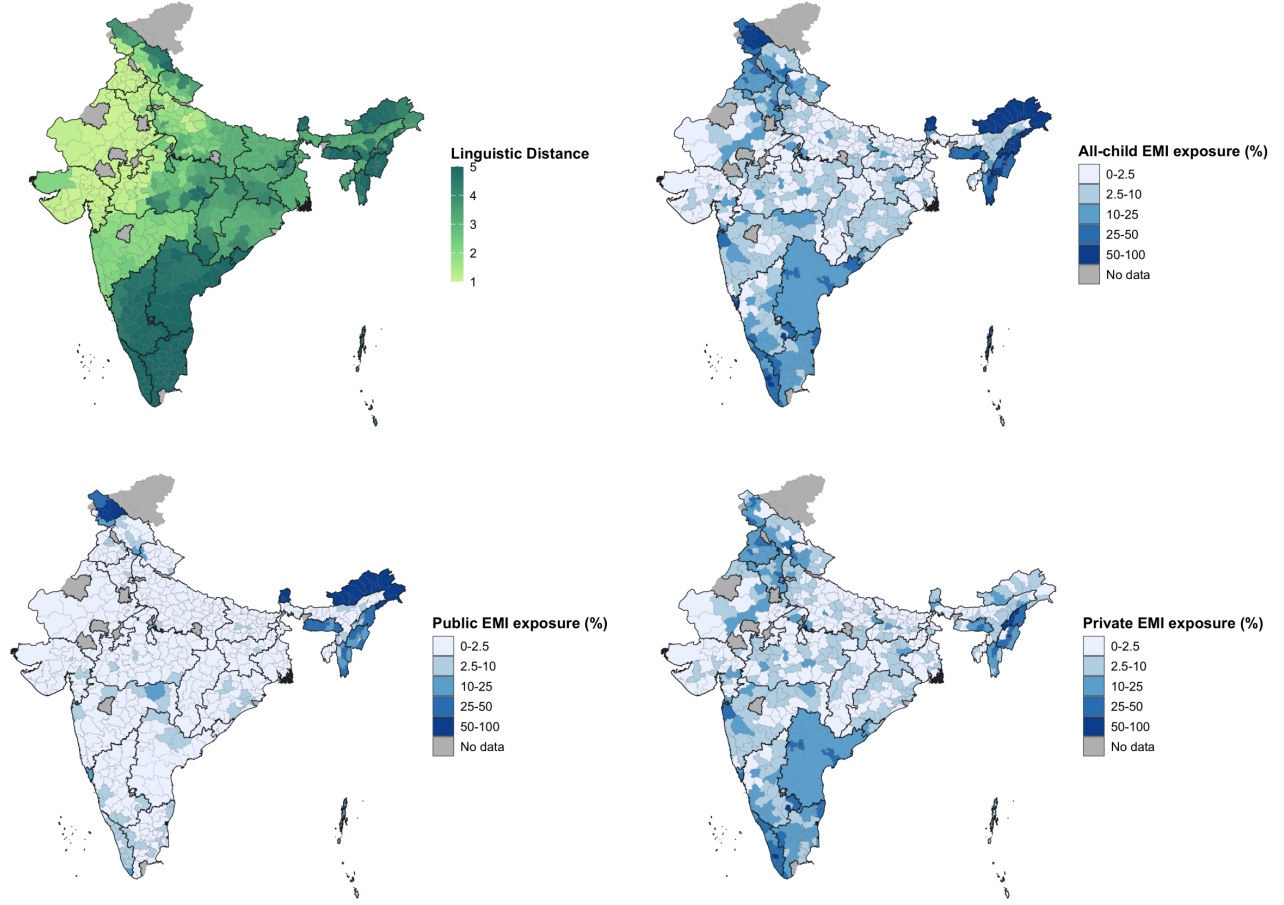


Figure 2: Linguistic distance from Hindi and English-medium schooling on common Census-2001 district geography. The three schooling panels use a common percentage scale. Sources: Census of India 2001 ([Office of the Registrar General and Census Commissioner of India 2001](#)) and NSS 64th Round, 2007–08 ([National Sample Survey Office 2011](#)).

5.2 English-Learning Opportunities as a Function of Institutional Factors

The schooling measures are organized very differently across states. Table~3 presents the same three geographic specifications for each schooling outcome: an unadjusted association, a specification with region fixed effects and predetermined Census-2001 controls, and a specification with state fixed effects and the same controls. The table reports standardized coefficients so that attenuation can be compared across outcomes, with state-clustered standard errors beneath each estimate. Nationally, linguistic distance is positively associated with every English-medium schooling margin in the table. The association attenuates sharply within states: the state-adjusted standardized coefficient is about 0.12 for enrollment, 0.03 for EMI among enrolled children, 0.06 for all-child EMI, 0.05 for public EMI, 0.09 for private EMI, and 0.05 for DISE English-medium enrollment.

The state-membership R^2 rows ask a separate descriptive question: how much district variation is associated with

state membership? State indicators explain roughly 84% of linguistic-distance variation. Across the schooling measures, they explain about 83–84% of all-child and conditional EMI, about 83% of DISE English-medium enrollment, and nearly 90% of public EMI, but only about 57% of private EMI and 36% of enrollment. These statistics locate the variation; they do not identify a causal effect of state institutions. Still, the public/private contrast is consistent with the institutional importance of state language policy. Indian state boundaries and schooling systems have historically been organized around language, and Jain (2017) shows that aligning official instructional language with local language after linguistic state reorganization substantially improved educational attainment.

Table 3: Linguistic Distance and English-Learning Opportunities across Geographic Specifications. Sources: NSS 64th Round, 2007-08; DISE 2007-08; Census of India 2001.

	(1)	(2)	(3)
<i>Enrollment</i>			
Linguistic distance from Hindi	0.249*** (0.079)	0.177*** (0.060)	0.120* (0.068)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State-membership R^2	0.359	0.359	0.359
<i>English-medium share among enrolled</i>			
Linguistic distance from Hindi	0.364*** (0.111)	0.175* (0.104)	0.033 (0.062)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State-membership R^2	0.840	0.840	0.840
<i>English-medium exposure among all children</i>			
Linguistic distance from Hindi	0.376*** (0.110)	0.190* (0.105)	0.063 (0.072)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State-membership R^2	0.829	0.829	0.829
<i>Public-school English-medium exposure</i>			
Linguistic distance from Hindi	0.331*** (0.127)	0.176* (0.096)	0.055 (0.077)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State-membership R^2	0.895	0.895	0.895
<i>Private-school English-medium exposure</i>			

Table 3: Linguistic Distance and English-Learning Opportunities across Geographic Specifications. Sources: NSS 64th Round, 2007-08; DISE 2007-08; Census of India 2001. (*continued*)

	(1)	(2)	(3)
Linguistic distance from Hindi	0.450*** (0.085)	0.190** (0.074)	0.093 (0.056)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	568	568	568
State-membership R^2	0.570	0.570	0.570
<i>DISE English-medium enrollment share</i>			
Linguistic distance from Hindi	0.403*** (0.111)	0.142* (0.085)	0.045 (0.053)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	520	520	520
State-membership R^2	0.841	0.841	0.841

Each outcome block reports standardized coefficients from district regressions on speaker-weighted linguistic distance from Hindi. State-clustered standard errors are in parentheses. Columns (1)-(3) are unadjusted, region fixed effects plus predetermined Census-2001 controls, and state fixed effects plus the same controls, respectively; the rows below each coefficient indicate these specification choices explicitly. Each outcome uses one fixed complete-case sample across its three specifications. State-membership R-squared is calculated from state indicators alone on the available state-coded support for each schooling measure, including DISE, and is descriptive rather than causal. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ Sources: NSS 64th Round, 2007-08; DISE 2007-08; Census of India 2001.

The NSS pattern is not an idiosyncrasy of household reporting. DISE English-medium enrollment and NSS EMI among enrolled children have a raw district correlation of 0.896, while DISE and all-child NSS EMI exposure correlate at 0.882. After I subtract each state’s mean from both measures, those correlations fall to 0.607 and 0.562, respectively. The decline is informative. A raw district correlation combines agreement across states with agreement among districts within the same state; state residualization removes the between-state component and asks only whether districts that are high relative to their own state are also high in the other source. Because the schooling geography above is strongly organized along state boundaries, removing that common state component should reduce the correlations if part of the cross-source agreement reflects broad state-level schooling systems. The remaining within-state correlations are still substantial, which makes simple NSS measurement error an implausible explanation for the weak within-state relationship between linguistic distance and EMI. The corresponding district-level NSS–DISE comparison is shown in Figure 5.

Neither EMI nor private schooling should be treated as a scalar measure of school quality. The project’s DISE indicators of pupil-teacher ratios, single-teacher schools, sanitation, private-sector composition, and EMI do not line up as one common quality dimension. External evidence points in the same direction. Private-school choice is often motivated by perceived quality as well as English (Joshi and Kumar 2017); large raw public/private

test-score differences can largely disappear under randomized school choice ([Muralidharan and Sundararaman 2015](#)); and stated EMI can differ markedly from actual classroom language ([Lahoti and Mukhopadhyay 2019](#)). The schooling variables here therefore describe access to different institutional options, not a ranking from low- to high-quality schools.

5.3 English-Learning Opportunities as a Function of Social Factors

The existence of options within a district does not imply equal access to them. Figure 3 compares Scheduled Tribe (ST), Scheduled Caste (SC), and Other Backward Class (OBC) children with children in the residual Other category within common districts. The social-group differences are present for enrollment but substantially larger for English-oriented and private schooling. Relative to Other children, mean district enrollment is lower by about 13.6 percentage points for ST children, 9.8 points for SC children, and 6.4 points for OBC children. EMI among enrolled children is lower by about 13.3, 17.2, and 13.2 points respectively. Private enrollment is lower by about 18.1 points for ST children, 23.5 for SC children, and 14.5 for OBC children. These differences are consistent with evidence that social institutions and household resources can shape who responds to English-oriented schooling opportunities ([Munshi and Rosenzweig 2006](#); [Endow 2019](#); [Raju 2023](#)).

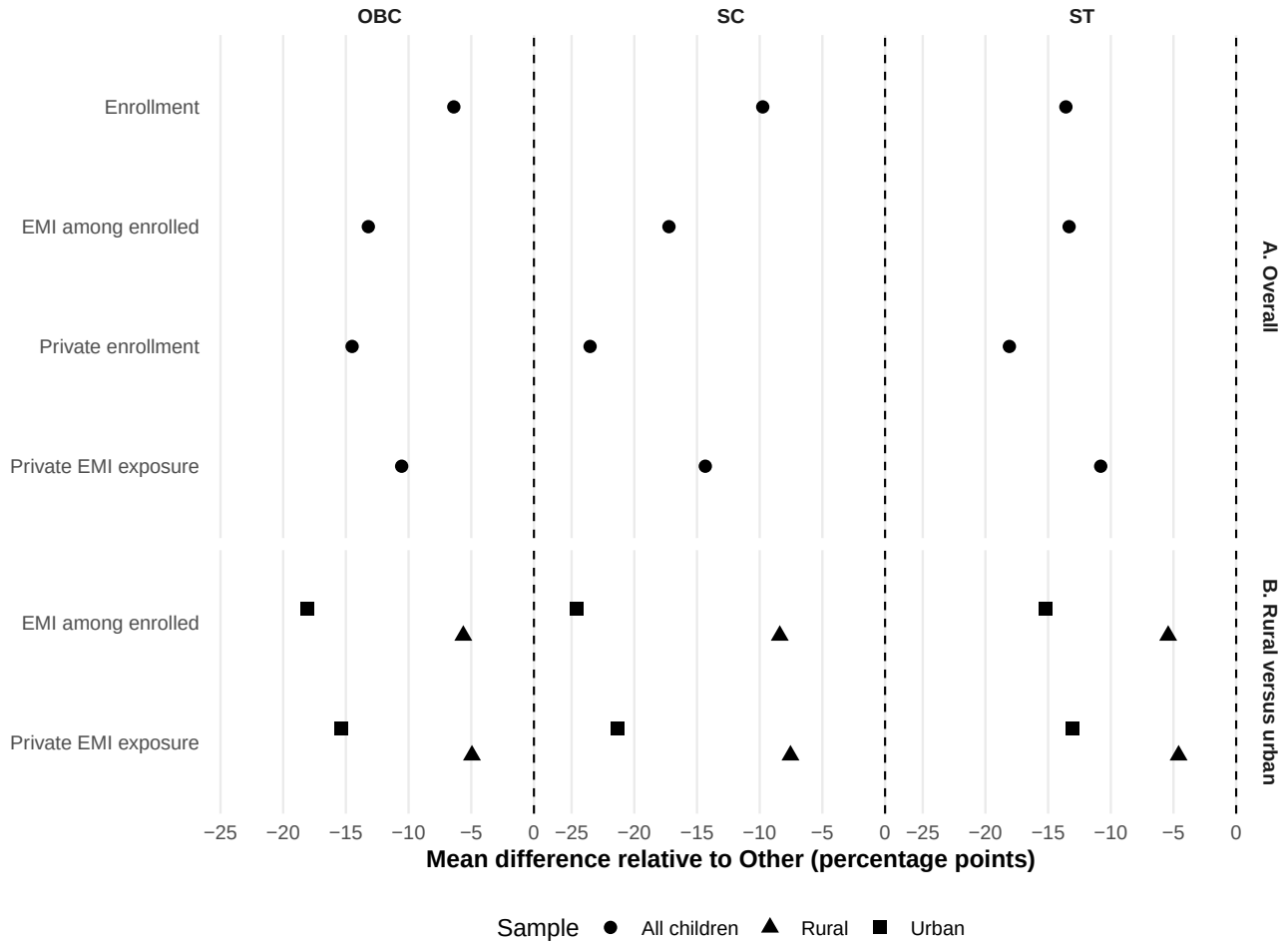


Figure 3: Social-group differences in access to English-oriented schooling. Panel A reports mean within-district percentage-point differences for OBC, SC, and ST children relative to the Other category in enrollment, EMI among enrolled children, private enrollment, and all-child private EMI exposure. Panel B separates rural and urban differences for EMI among enrolled children and private EMI exposure. Source: NSS 64th Round, 2007–08 ([National Sample Survey Office 2011](#)).

The rural/urban comparison makes the social stratification more pronounced. For SC children, for example, the difference in EMI among enrolled children is about 8.4 percentage points in rural districts and 24.6 points in urban districts; the corresponding difference in private EMI exposure is about 7.5 and 21.4 points. OBC and ST children show the same qualitative pattern. A plausible interpretation is that more differentiated urban schooling markets offer a wider range of formal options while household resources and social position still condition which options are reachable. I treat that as a descriptive sorting pattern rather than evidence that urbanization itself causes the larger differences.

The social-group differences are not mainly a male-versus-female phenomenon in these data: the corresponding

decompositions for boys and girls are directionally similar. That does not imply that sex is unimportant for schooling choice. Munshi and Rosenzweig (2006) document sharply gendered responses to English schooling in Bombay, and government programs operating before the NSS round explicitly targeted girls’ education (Press Information Bureau 2008). In this national cross-section, however, social group and rural/urban setting distinguish access more strongly than the sex split.

The result is a constraint on how the schooling menu should be interpreted. A district can contain English-medium and private schools without placing those options within equal reach of every household. The next section asks whether these observed English-learning opportunities are associated with the later economic outcomes that might make households want them in the first place.

6 Anticipated Economic Returns to English Learning

6.1 Economic Outcomes as a Function of English Learning

English-language skills are a form of human capital (Chiswick and Miller 1995; Azam et al. 2013), and evidence from India shows why households may value them. Using nationally representative 2005 household data, Azam et al. (2013) find English fluency associated with a 34% hourly wage premium for men and a 22% premium for women; even workers with a little English receive smaller positive premia, and the returns are greater for more educated workers. Refeque and Azad (2022) later documents substantial English-skill earnings associations through 2011–12. The strongest evidence for a causal effect on wages comes from Chakraborty and Bakshi (2016), who exploit West Bengal’s withdrawal of English from primary schools and estimate substantially lower later wages for cohorts more exposed to the policy. These studies motivate the payoff side of the household investment problem.

I do not observe households’ beliefs about those payoffs, and I do not assume that parents choosing schooling in 2007–08 foresaw district consumption in 2022 or 2023. I instead use an **adaptive expectations** interpretation: households update beliefs using outcomes they have observed rather than possessing perfect forecasts of future conditions (Nerlove 1958). The external wage literature establishes that English skills had economic value around the period in which the schooling choices studied here were being made. I then use the later district outcomes only to characterize how English-intensive schooling was eventually associated with local living standards. Those later associations are realized local outcomes, not estimates of the beliefs households actually held.

Table~4 compares five schooling margins on common district support. Each entry comes from a separate district regression with state fixed effects and predetermined Census-2001 controls; the table assembles those regressions so that the schooling margins can be compared on the same scale. In the 2022–23 ANCOVA specification, a 10

percentage-point increase in enrollment is associated with only about 0.7% higher real mean consumption and is imprecisely estimated. The corresponding associations are about 3.9% for EMI among enrolled children, 5.1% for all-child EMI, 4.1% for public EMI, and 5.3% for private EMI. The 2004–05 to 2022–23 long-change specification particularly singles out private EMI, for which the association is about 5.4% per 10 percentage points. The analogous 2023–24 estimates are smaller, and none survives the predeclared family correction.

Table 4: English-Intensive Schooling, Later Consumption, and Predetermined Complements. Sources: NSS 64th Round, 2007-08; Census of India 2001; NSS 61st Round, 2004-05; HCES 2022-23 and 2023-24.

	(1)	(2)	(3)	(4)
Enrollment	0.70 (1.70)	1.27 (1.44)	-0.02 (1.08)	0.72 (1.44)
EMI among enrolled	3.88** (1.15)	2.79 (1.13)	2.31 (1.16)	0.82 (1.23)
All-child EMI	5.06*** (1.40)	3.88** (1.23)	2.62 (1.45)	1.06 (1.62)
Public EMI	4.07* (1.48)	1.13 (2.20)	2.04 (1.18)	-0.94 (2.49)
Private EMI	5.31** (1.69)	5.44*** (1.29)	2.94 (1.98)	2.53 (1.79)
All-child EMI × Baseline human capital		1.50*** (0.45)		
All-child EMI × Urbanization		1.83** (0.61)		
All-child EMI × ST concentration		0.04 (0.52)		
Private EMI × Baseline human capital		1.47 (0.80)		
Private EMI × Urbanization		2.00 (0.98)		
Private EMI × ST concentration		0.40 (0.83)		
Endpoint	HCES 2022-23	HCES 2022-23	HCES 2023-24	HCES 2023-24
Estimand	ANCOVA	2004-05 to 2022-23 change	ANCOVA	2004-05 to 2023-24 change
State fixed effects	Yes	Yes	Yes	Yes
Predetermined controls	Yes	Yes	Yes	Yes
Observations	524	524	522	522

Each schooling-margin coefficient comes from a separate district regression and reports the percent difference in real mean MPCE associated with a 10 percentage-point increase in that schooling margin; coefficients shown in the same column are assembled for comparison rather than jointly estimated. Columns use state fixed effects, predetermined Census-2001 controls, and common schooling support within each endpoint/estimand. State-clustered standard errors are in parentheses; stars use Holm-adjusted p-values within the predeclared schooling-welfare family. The six interaction rows appear only for the 2004-05 to 2022-23 long change and report how the all-child or private-EMI association differs with a one-standard-deviation increase in predetermined human capital, urbanization, or ST concentration. Those rows also come from separate registered regressions and use state-clustered standard errors and Holm-adjusted p-values within the predeclared complement family. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. These are descriptive conditional associations, not causal returns or treatment-effect heterogeneity. Sources are identified in the caption.

The comparison is economically suggestive without being a causal return to EMI. English-oriented schooling is socially selected, and the district coefficients need not equal individual wage returns. This caution is especially important for private schooling: experimental school-choice evidence shows that large raw private-school advantages need not persist under random assignment (Muralidharan and Sundararaman 2015). Still, the contrast gives economic content to household demand for English-oriented schooling: the English-intensive margins are much more strongly associated with later local living standards than enrollment alone. This is also consistent with Azam et al. (2013)’s evidence that English skills and education are complementary in the labor market.

6.2 Complements to English Learning

The same Table~4 asks whether the 2004–05 to 2022–23 association is stronger where complementary local capacity was already greater before the schooling round. The interaction rows standardize each complement so that they report how much the schooling association changes with a one-standard-deviation increase in the predetermined characteristic. For all-child EMI, a one-standard-deviation increase in baseline human capital raises the association by about 1.5 percentage points, while a one-standard-deviation increase in urbanization raises it by about 1.8 points; both survive the predeclared family correction. ST concentration does not. The private-EMI interactions with human capital and urbanization have the same sign, but neither survives its family correction.

These results suggest that access to English-intensive schooling is not by itself sufficient for equal economic conversion. I use **complement** in its ordinary economic sense here: a pre-existing local characteristic is complementary when the schooling–welfare association is larger where that characteristic is more abundant. Human capital and urbanization fit that description for all-child EMI. A separate predeclared interaction with 2005 IT-employment intensity does not: neither the all-child EMI interaction nor the linguistic-distance interaction provides evidence that the results are simply a feature of districts that already had high IT-employment intensity. The relevant local capacity therefore appears broader than initial IT-sector concentration.

This qualification matters for the paper’s inequality argument. The schooling margins most associated with later prosperity are already unequally accessed, and the association is strongest in places with greater pre-existing human-capital and urban capacity. A household can therefore face a high-value English opportunity while living in a local economy with fewer of the complementary conditions under which that opportunity is associated with later economic advantage.

6.3 Economic Outcomes as a Function of Linguistic Conditions

The third part of the returns question concerns inherited linguistic conditions themselves. Linguistic distance does not closely track formal EMI within states, so I ask whether it is nevertheless associated with later economic outcomes outside the measured schooling margin.

The timing is informative. Figure 4 follows the coefficient on linguistic distance across the registered later-consumption endpoints. The 2009–10 and 2011–12 estimates are small and imprecise. Positive long-change associations appear much later, especially from 2004–05 to 2022–23, and remain positive but less precise in 2023–24. This timing is consistent with slow human-capital accumulation, migration, or structural adjustment, but it does not identify which process accounts for the delayed association.

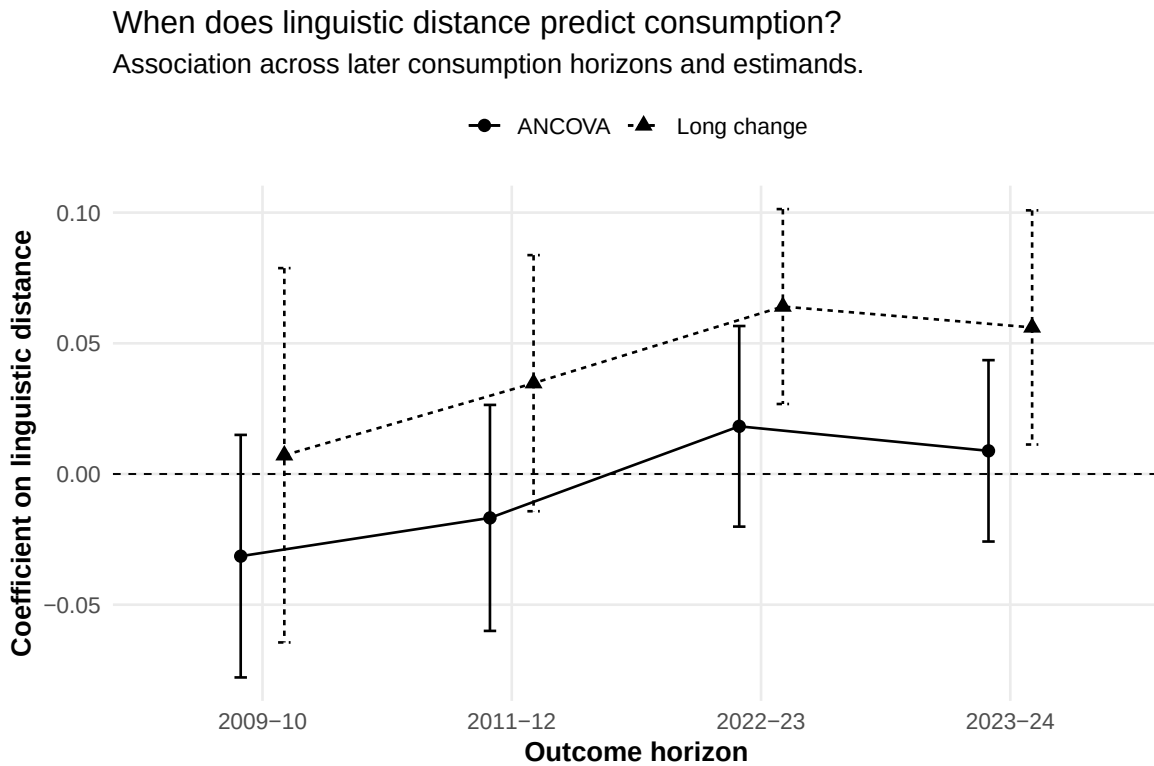


Figure 4: Association between linguistic distance and real mean household consumption across later horizons. Points show ANCOVA and long-change coefficients with 95% state-clustered Wald confidence intervals; all specifications include state fixed effects and predetermined Census-2001 controls. Sources: Census of India 2001; NSS 61st, 66th, and 68th Rounds; HCES 2022–23 and 2023–24 ([Office of the Registrar General and Census Commissioner of India 2001](#); [National Sample Survey Office 2007, 2012b, 2014, 2024, 2025a](#)).

Table~5 places that long-run consumption pattern beside other local outcomes without selecting only significant estimates. Greater linguistic distance is associated with deeper household literacy and graduate access, a higher

skilled share among recent work in-migrants, larger gains in banking access and television ownership, and a shift in employment composition toward services and away from manufacturing. The migration result is about the *skill composition of in-migrants at their destination*, not the number leaving a district or the interstate share of migration. The latter is null.

Table 5: Linguistic Conditions and Selective Local Development. Sources:
Census of India 2001 and 2011; Economic Census 2005 and 2013; NSS
66th Round; PLFS 2017-18.

	(1)	(2)	(3)	(4)
<i>Panel A: Household capacity and assets</i>				
Linguistic distance from Hindi	0.0092**	0.0041**	0.0475***	0.0141**
	(0.0035)	(0.0015)	(0.0113)	(0.0050)
Dependent variable	2+ literates	Graduate access	Banking access	Television ownership
Period	2001-11	2001-11	2001-11	2001-11
Fixed effects	State	State	State	State
Predetermined controls	Yes	Yes	Yes	Yes
Observations	355	355	355	355
<i>Panel B: Migration and sectoral composition</i>				
Linguistic distance from Hindi	0.0129**	-0.0023	0.0206**	-0.0179**
	(0.0044)	(0.0166)	(0.0066)	(0.0062)
Dependent variable	Skilled recent-work migrant share	Interstate migrant share	Services employment share	Manufacturing employment share
Period	2011	2011	2005-13	2005-13
Fixed effects	State	State	Region	Region
Predetermined controls	Yes	Yes	Yes	Yes
Observations	355	355	354	354
<i>Panel C: Employment scale and labor outcomes</i>				
Linguistic distance from Hindi	-0.0226	-0.0074	-0.0085	
	(0.0220)	(0.0057)	(0.0057)	
Dependent variable	Log nonfarm employment	Labor-force participation	Employment rate	
Period	2005-13	2017-18	2017-18	
Fixed effects	Region	State	State	
Predetermined controls	Yes	Yes	Yes	
Observations	354	467	467	

Each column is a separate district regression. The coefficient reports the association of one Shastri linguistic-distance degree with the listed outcome in its native unit; state-clustered standard errors are in parentheses and stars use Holm-adjusted p-values within the registered outcome/specification family. Household-capacity, migration, finance, asset, and labor regressions use state fixed effects plus predetermined controls; Economic Census regressions use region fixed effects plus the same controls, as reported in the table. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The 2001-11 Census and 2005-13 Economic Census windows overlap the 2007 schooling measurement, so these are co-evolving development margins rather than post-treatment EMI mechanisms. PLFS 2017-18 supplies the displayed labor endpoint; the semantic CSV retains the raw and Holm-adjusted p-values and matched NSS66 adjusted p-values. Heterogeneous outcomes remain in native units, and null outcomes are retained by design. Sources are identified in the caption.

The absences help define the pattern. Nonfarm employment growth is not robustly positive, and neither labor-force participation nor the employment rate shows a broad positive association. The wider welfare-definition family also produces no family-wide signal, so the preferred real-mean-consumption result should not be read as universal improvement in every welfare measure. The Economic Census evidence is similarly compositional: services rise and manufacturing falls, while total establishment and employment growth do not show a comparably robust expansion.¹

This combination resembles a broader feature of India’s development path. Kochhar et al. (2006) emphasize the country’s service- and skill-intensive pattern relative to the manufacturing-led path followed by several other fast-growing Asian economies. More recent work also stresses how tradeable services can become increasingly important to development as technology changes the tradability of service work (Baldwin and Forslid 2023). My estimates do not show that linguistic distance caused such structural transformation. They show that the inherited linguistic gradient is associated with a selective set of human-capital, skill-sorting, financial, and sector-composition outcomes rather than with generalized employment expansion.

Timing constrains the interpretation further. The Census household and housing outcomes cover 2001–11, while the Economic Census changes cover 2005–13; both windows overlap the 2007–08 schooling measurement. I therefore treat these as co-evolving local conditions, not as downstream mediators of measured EMI. Taken together, the evidence suggests that anticipated returns to English learning depend not only on the value of English skills themselves, but also on the local capacity through which those skills can be converted into economic opportunities.

7 Conclusion

English learning is often presented as a way for households to loosen the connection between children’s prospects and the circumstances into which they are born (Itani et al. 2015; Bedi 2019). The premise is economically plausible: English proficiency carries substantial labor-market rewards in India (Azam et al. 2013; Refeque and Azad 2022), and English schooling has been associated with access to occupations outside traditional caste-linked networks (Munshi and Rosenzweig 2006). I began from a narrower question. If English learning is to expand opportunity, what conditions surround a household’s decision to invest in it, and are those conditions themselves related to inherited and social inequality?

The evidence points to inequality in each of the three conditions studied in this paper. Local linguistic conditions affect the relative cost of acquiring a bridge language and persist across generations. Their relationship with

¹The sector-composition results are strongest with region fixed effects: the raw state-clustered p-values are 0.002 for services and 0.004 for manufacturing. Under state fixed effects the corresponding raw p-values are 0.031 and 0.027, but both adjust to about 0.110 under the predeclared family correction.

actual language acquisition depends on social and institutional context, however. Formal English schooling is more institutionally organized than linguistic incentives alone would imply: state systems account for much of its geographic variation, while access to English-medium and private schooling remains strongly stratified by social position within the same districts. The local availability of a school option therefore does not imply that every household faces the same effective choice set.

The same distinction carries into the payoff side of the household’s decision. English-intensive schooling is more strongly associated with later local consumption than enrollment alone, but that association is stronger where baseline human capital and urbanization are greater. The inherited linguistic gradient also accompanies a broader and selective pattern of local change in human capital, skilled in-migration, financial access, and sectoral composition rather than a general expansion of employment. India’s service- and skill-intensive development path provides useful context for this pattern (Kochhar et al. 2006; Baldwin and Forslid 2023). These are descriptive district relationships, not estimates of individual returns or evidence of a single causal sequence running from language to schooling to development.

The attempted instrumental-variable design reinforces that distinction. Linguistic distance predicts English-medium schooling across India but supplies little within-state variation under the preferred specification, and reasonable alternative language measures do not restore a persuasive first stage. I therefore cannot use linguistic distance to identify the causal effect of English-medium instruction. More importantly for the interpretation of the paper, inherited linguistic conditions, formal schooling choices, and local economic change should not be treated as interchangeable measures of one underlying process. English-medium schooling is one observed means through which children may acquire English skills; it is not a sufficient measure of the economic opportunity associated with English.

A related distinction concerns what English-medium instruction can be expected to accomplish. The economic value of learning English is not the same as the pedagogical value of learning every school subject through English. Evidence from India shows substantial educational costs when the language of instruction is unfamiliar to students (Jain 2017; Nair 2015; Mohanty 2022). I do not identify the causal effect of English-medium instruction on learning, and district outcomes do not identify the later welfare of the same children observed in the schooling data. Migration and changing district populations further separate local development from individual returns.

English skills are economically valuable, and households have reason to invest in them. But investment does not begin from an equal position. A child can face high prospective returns to English while inheriting a less favorable linguistic starting point, facing a more restricted effective schooling choice set, or living in a local economy with fewer complementary resources. A high-return educational opportunity therefore need not equalize opportunity when entry into it—and the resources needed to benefit from it—are themselves unequally distributed.

8 Technical Note for Replication

All reported tables and figures are generated from registered build targets. The analysis is harmonized to Census-2001 districts; the main manuscript reports bounded summaries, while construction (Section [A](#)), migration (Section [B](#)), selection (Section [C](#)), and weak-identification (Section [D](#)) material is retained in the appendices. More extensive validation and local-development results remain available in machine-readable analysis outputs rather than as separate manuscript appendices. Reproduction instructions, data-availability constraints, and source documentation are provided in `REPLICATION.md`, `DATA_AVAILABILITY.md`, and the repository documentation.

A Major Data Contributions

Several of the datasets used in this paper are not directly comparable across time or source. District boundaries change; survey district codes follow different vintages; archived school records use their own district labels; and consumption surveys span multiple geographic classifications. I therefore treat geographic and measurement harmonization as part of the empirical work rather than as a preliminary name-matching exercise. This appendix describes the constructions that are most important for interpreting and reproducing the paper.

A.1 District Lineage Harmonization

District names are not longitudinal identifiers in India. Between the 1961 and 2001 Censuses the number of districts rose from 339 to 593, through renamings, partitions, mergers, and boundary changes (Kumar and Somanathan 2016). The 2011 Census contained 640 districts after another 47 were created during the intervening decade (Office of the Registrar General and Census Commissioner, India 2012). Later surveys and the 2022–24 Household Consumption Expenditure Surveys therefore cannot simply be joined to the 2001 Census by district name. The difficulty is substantive: if a later district contains territory from several earlier districts, assigning it to whichever earlier district has the most similar name changes the population represented by the observation.

I use the 593 Census-2001 districts as the reference geography. Historical boundary evidence extends back to 1961, while source-specific links carry later surveys through the 2023–24 consumption round. This choice keeps the linguistic measures and predetermined Census controls on the geography in which they are observed and gives every later source a common target. Kumar and Somanathan’s district histories provide a benchmark for pre-2001 changes (Kumar and Somanathan 2016). For later changes I combine Census district records with the Ministry of Panchayati Raj’s Local Government Directory (LGD), which assigns location codes and records the creation and reorganization of local government units (Ministry of Panchayati Raj, Government of India 2026). I cross-check these records against the India State and District Evolution Database, the India District Changes Tracker, and published survey concordances (Jolad et al. 2026; Jaacks et al. 2020; Deshpande et al. 2024). SHRUG locality keys supply an independent, population-based account of how 2001 and 2011 places overlap (Asher et al. 2021). Table 6 summarizes the roles of these source families.

The current procedure supersedes the fuzzy-join method described in the original draft. I standardize spelling and names to generate possible matches, but string similarity cannot accept a district identity. Each survey district must first be tied to a registered Census or LGD unit using its code, an exact or documented renamed identity, or a reviewed source record. Candidate matches that require judgment enter a review ledger that records the decision and the evidence used. The lineage checks require every accepted decision to cite a registered source; in the current build all 1,254 accepted source identities satisfy that requirement. Reorganizations are stored

Table 6

District-Lineage Sources and Roles. Sources: Census of India; Ministry of Panchayati Raj Local Government Directory; Kumar and Somanathan (2016); India State and District Evolution Database; India District Changes Tracker; Development Data Lab SHRUG; Deshpande, Khanna, and Walia concordances.

Source	Coverage	Role
Census and NSS district records	Census 2001 and 2011; NSS 64th and 75th rounds	Define reference districts and resolve survey identities
Local Government Directory	2001 onward	Verify district creation, renaming, and Census-code continuity
Kumar and Somanathan district histories	1961-2001	Benchmark historical partitions and population transfers
India State Stories / District Changes Tracker	1951 onward / 2001-2020	Cross-check district histories and predecessor relationships
SHRUG locality and district keys	Population Censuses 1991-2011	Measure locality-based overlap when later districts cross 2001 boundaries
Published survey concordances	Census, NSS, and PLFS district codes	Cross-check survey-to-Census district links

Note: Rows summarize source families rather than individual files. Detailed source IDs and adjudication ledgers remain available in the lineage CSV outputs.

as dated predecessor–successor relationships. A split links one predecessor to several successors, a merger links several predecessors to one successor, and a documented rename or unchanged district preserves identity. Each reviewed relationship records the source used to support it.

The evidence sources have different roles rather than equal votes. Census records and LGD codes provide the strongest evidence on official district identity and code continuity. Published district histories and survey concordances corroborate those identities and describe reorganizations. SHRUG locality keys are used when population overlap is needed to quantify how a later district relates to 2001 geography. A manual review cannot convert a candidate into an accepted link without citing one of the registered sources summarized in Table 6.

I then trace each accepted source unit to its Census-2001 predecessor. Three cases are distinguished. First, an unchanged district or documented rename receives a unit weight on the corresponding 2001 district. Second, a later district created wholly from one 2001 district can be assigned to that predecessor when the district history establishes a nested, single-parent relationship, meaning that the later district lies wholly within one 2001 predecessor. Third, a later district can overlap more than one 2001 district. These multi-parent cases do not enter the preferred panel. For sensitivity analysis only, I use SHRUG locality populations to calculate the share of the later district attributable to each 2001 predecessor; accepted fractional weights must be nonnegative and sum to one within the source district. This treatment follows the same principle as the population weights used by Kumar and Somanathan (2016) for historical district partitions: aggregation should represent territory and population, not similarity between district names.

The **conservative** lineage variant admits only direct identities, documented renames, and deterministic one-

parent transitions. The **primary** variant additionally admits reviewed single-parent cases supported either by official Census/LGD code continuity or by independent district-history evidence corroborated with SHRUG locality overlap; for the latter cases, at least 99 percent of the source district’s population must be represented in the locality crosswalk. The **full reviewed** variant is a sensitivity panel that also admits accepted fractional allocations across multiple 2001 predecessors. For the two schooling waves, these rules yield 427 common districts in the conservative panel, 573 in the primary panel used in the main analysis, and 587 in the full reviewed panel. Multi-parent cases remain outside the primary panel even when a fractional sensitivity allocation is available.

The same principle governs aggregation. Counts are summed across source units assigned to a common 2001 district. Rates, shares, and means are reconstructed from pooled numerators and denominators rather than averaged across successor districts. Distributional measures that cannot be recovered from sufficient statistics, such as the consumption Gini, are reconstructed from pooled household records where the underlying microdata permit it. This distinction prevents a small successor district from receiving the same weight as a much larger one merely because both descend from the same reference district.

Geometry is treated separately from statistical allocation. The main geographic files use Census-2001 district boundaries. Where the available 2001 shapefile omits a district whose boundary is demonstrably unchanged, I carry back a 2011 polygon only when the Census administrative atlases establish exact continuity ([Office of the Registrar General and Census Commissioner, India 2001, 2012](#)); the current build uses this rule for eleven districts. I do not infer missing boundaries by reallocating area between polygons for these geometry repairs. The statistical crosswalk and the geographic display therefore share district identities without allowing the availability of a convenient polygon to determine how observations are allocated.

The purpose of these rules is not to maximize the number of matched districts. It is to make the meaning of a district observation stable enough that differences across years reflect changes in measured outcomes rather than undocumented changes in territory. The three lineage variants make the remaining coverage trade-off explicit. The validation appendix reports the corresponding sample coverage and shows that the paper’s identification conclusion is not a consequence of choosing the primary rather than the conservative or full reviewed lineage.

A.2 DISE Data Reconstruction

The District Information System for Education (DISE) is valuable here because it records schools rather than household reports of schooling. The historical files are also difficult to use directly. The original publication site is no longer maintained, file formats change across years, and the raw enrollment workbooks encode medium of instruction in numbered slots whose language meaning can differ across districts. The surviving publication index maintained by Arun Mehta identifies the district report cards and raw files released by NIEPA/NUEPA and notes

that 2005–06 was the first year in which DISE covered the entire country using a common data-capture format (Mehta 2022). I used that index to locate an Internet Archive snapshot of the former DISE district-report-card site and recovered the corresponding workbooks and published report cards (District Information System for Education 2017). The local archive spans 2002–03 through 2015–16; the 2005–06 through 2007–08 files provide the baseline measures used in the paper, while later years are retained for longitudinal checks.

I record each academic year in a source registry that specifies the archived raw file, any file contained inside a ZIP archive, the enrollment, school, and teacher sheets to read, the corresponding report-card volumes, and any known source problem. This is necessary because neither the workbook layout nor the file extension reliably identifies where each measure is stored. Each year is therefore read using its registered sheet structure rather than a single layout assumed across all vintages. Earlier 2002–05 files are retained as inventory; the baseline construction begins in 2005–06, when nationwide coverage becomes comparable (Mehta 2022).

The most consequential reconstruction concerns medium of instruction. In the 2005–08 raw enrollment sheets, up to five medium slots contain grade-specific pupil counts, but the slot number is not itself a language code. I sum the grade cells within each slot and identify the slot’s language from the corresponding published district report card. The resulting crosswalk contains 7,211 district-year-slot identities, each tied to a report-card PDF and page. English enrollment is the sum of the slots identified as English; Hindi enrollment is constructed in the same way. I do not turn an unidentified positive slot into zero English enrollment. If the English slot itself can be identified, its observed pupils remain usable; if a positive unidentified slot means that the absence of English cannot be established, the English count is left missing. This rule prevents uncertainty about a report label from being converted mechanically into zero exposure.

The denominator is reconstructed separately from the medium slots. For the baseline years, I use total enrollment in Classes I–VIII obtained from the grade counts when available, followed by the workbook’s direct enrollment total and then the sum of government and private enrollment when the earlier measures are unavailable. Medium-classified enrollment is used only as a consistency check; it is never substituted for total elementary enrollment. English-medium enrollment share is therefore English-medium pupils divided by total elementary enrollment, not the share of pupils whose medium happened to be classified in the source file. I also sum government and private management categories separately to construct private enrollment and private-school shares. School and teacher files supply the corresponding school counts, teacher counts, single-teacher-school counts, and girls’-toilet counts; ratios are computed only after these underlying counts have been assembled.

For measures scaled by the school-age population rather than by observed enrollment, I use Census counts of children ages 6–13. I first place the 2011 age counts on the Census-2001 district geography using only deterministic district links, then combine the 2001 and 2011 anchors to estimate each district’s annual log population growth.

Population in an intervening academic year is obtained by log-linear interpolation between those anchors. The English-medium gross enrollment ratio divides English-medium pupils by this interpolated age-6–13 population. Post-2011 values used only in longitudinal checks extend the same growth rate beyond 2011; the paper’s 2005–08 baseline measures rely on interpolation rather than extrapolation.

District harmonization follows the lineage rules described in Section A.1, but DISE school counts impose a stricter restriction. A DISE district is carried to the Census-2001 geography only when it has an exact identity or a reviewed one-to-one lineage link with weight one. Fractional allocations across several predecessors are excluded because splitting reported school or pupil counts by population weights would create fractional source counts that were never observed. When several source districts correspond deterministically to one Census-2001 district, I sum the pupil, school, and teacher counts first and recompute shares and ratios from the pooled numerators and denominators.

Published report cards are used both to decode the raw files and to check selected reconstructed cells. As a direct parser check, three registered 2005–06 medium-of-instruction cells for Kupwara reproduce the published report exactly, and the build fails if those cells cease to match. For later years, published report-card pages provide English and Hindi pupil counts when the workbook no longer supplies stable slot identities. The 2009–10 archive is a special case: the second report-card volume was not preserved, so I use the archived complete *District Elementary Education Report Card: 2009–10* where the missing volume is needed ([National University of Educational Planning and Administration 2012](#)). In 2010–11 the raw enrollment workbook contains a documented header/data alignment problem; I therefore use the published current-year district total when it can be matched and leave the denominator unavailable otherwise rather than fall back to the corrupted raw total. These source-specific rules are registered by year instead of being inferred after looking at regression results.

The independent NSS comparison provides a useful check on the resulting baseline measure. Figure 5 plots 2007–08 DISE English-medium enrollment against NSS English-medium instruction among enrolled children. The raw district correlation is 0.896. After subtracting state means from both series, the correlation remains 0.607. The decline is expected because both schooling measures have a strong state component, while the remaining association shows that the two independent sources agree materially even within states.

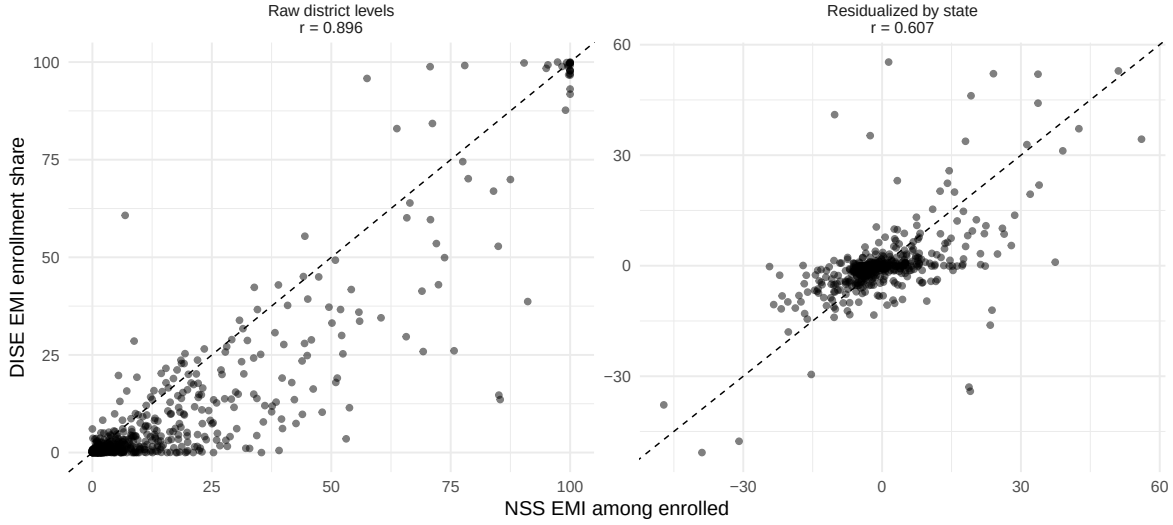


Figure 5: Independent agreement between NSS and DISE English-medium measures. The left panel compares district levels; the right panel subtracts each source’s state mean before comparison. Sources: NSS 64th Round, 2007–08 ([National Sample Survey Office 2011](#)) and DISE District Report Cards, 2007–08 ([District Information System for Education 2017](#)).

A.3 Linguistic Measures

The linguistic measures are intended to capture one specific economic margin: how costly it is to acquire Hindi, relative to English, as a language of wider communication. They are not measures of English proficiency, school quality, or a child’s complete language environment. I follow Shastry (2012) in using linguistic distance from Hindi because her argument is comparative. As the distance between one’s mother tongue and Hindi increases, I expect the opportunity cost of learning English over Hindi to decrease. The primary district measure therefore begins with language-level distances from Hindi and then combines them with the local distribution of mother-tongue speakers.

The speaker distribution comes from the mutually exclusive mother-tongue rows of Census 2001 Table C-16 ([Office of the Registrar General and Census Commissioner of India 2001](#)). Census language-group subtotals are not treated as additional speakers. This matters particularly for the Census Hindi group, which contains distinct mother tongues such as Bhojpuri and Rajasthani varieties. I assign distance at the mother-tongue level first; a distinct mother tongue cannot inherit Hindi’s zero merely because the Census places it beneath the Hindi group.

The preferred language scores use Shastry’s ordinal 0–5 classification ([Shastry 2012](#)). Hindi and Urdu are the zero-distance reference. Published Indo-European languages retain Shastry’s reported degree, while languages outside Indo-European receive degree five only when their family identity has been reviewed. Native English is treated separately: it is neither a zero-distance Hindi language nor part of the weighted-distance mean. The primary

speaker-weighted distance from Hindi is the Census-speaker-weighted mean among mapped mother tongues with positive Shastry distance. This differs from the older **top-three-language distance**, which retains only the three largest mapped mother tongues in each district. The primary construction uses the full mapped mother-tongue distribution and therefore avoids making the district measure depend on an arbitrary three-language cutoff. I retain the top-three measure only for comparison with the earlier analysis.

Not every Census mother tongue has an unambiguous published Shastry category. I therefore use a reviewed crosswalk rather than assigning degrees from names alone. The published Shastry concordance is applied first. For finer Indo-European mother tongues absent from that concordance, reviewed decisions use historical classification evidence from the *Linguistic Survey of India* (Grierson 1903–1928) and a named published Shastry anchor. Lexical comparisons from Kogan (2017) and ASJP v21 (Wichmann et al. 2025) are used only when they help distinguish nearby classifications; they are not averaged into the preferred distance. The current frozen review ledger contains both accepted assignments and explicitly unresolved rows. A mother tongue for which the evidence does not identify one defensible degree remains unmapped in the preferred construction rather than being assigned the degree that produces greater statistical relevance.

Unmapped speaker mass is retained explicitly. For the preferred weighted mean, only mapped positive-distance speakers enter the numerator and denominator, and the mapped-speaker share is reported separately. For population-share measures, unresolved and native-English speakers remain in the district population denominator. Thus the **share speaking languages distant from Hindi** is the percentage of all mother-tongue speakers whose mapped language is at least three Shastry degrees from Hindi. The five positive-distance shares similarly record the percentage of district speaker mass at each degree from one through five; when those shares are entered jointly, distance zero is the omitted reference category. I also retain low- and high-degree assignments recorded for reviewed ambiguous languages as bounding sensitivity checks, but those alternatives never replace the frozen preferred assignments.

The 1991 reconstruction applies the same conceptual rules to the *Language Atlas of India 1991*, which reports the geographic distribution of 114 classified languages (Office of the Registrar General and Census Commissioner, India 2004). Each Atlas language is linked to the same Shastry basis, and accepted district-language counts are checked against the Atlas population totals. Because unresolved cells can alter a weighted mean even when they represent little population, I calculate bounds that assign unresolved nonzero language mass to the lowest and highest feasible Shastry degrees. The preferred historical series requires at least 99 percent accepted speaker coverage and a resulting distance-bound width no greater than 0.5 degree before a district is retained. The resulting 1991 districts are then placed on Census-2001 geography using the lineage rules in Section A.1. This is the historical analogue used for persistence and source validation, not a separate preferred modern measure.

Two independent alternatives deliberately change what linguistic distance means. The **genealogical distance from Hindi** uses the reviewed Census-to-Glottolog 5.3 crosswalk and counts separation along Glottolog’s language-family genealogy (Hammarström et al. 2026). Glottolog provides stable language identifiers and an explicit genealogical classification; I do not rescale this edge count onto Shastry’s 0–5 scale. The **lexical noncognacy with Hindi** measure instead uses the Dyen–Kruskal–Black Indo-European cognate data (Dyen et al. 1992; “Comparative Indo-European Database” 1997). At the language level it is one hundred minus the percentage of cognate items shared with Hindi. Following the convention described by Shastry, reviewed non-Indo-European languages outside the Dyen comparison receive five percent shared words, or a noncognacy value of 95; unresolved Indo-European mother tongues remain unresolved. Kogan and ASJP evidence are not numerically substituted for missing Dyen observations. These alternatives are robustness measures because they rely on different linguistic information, not competing versions from which I select the strongest empirical result. Table 7 summarizes the manuscript-facing constructions and their roles. The distinctions in the final column are substantive: Shastry distance is the primary comparative-cost measure; the top-three construction is historical comparability; the distant share and distance composition capture nonlinearity; the 1991 series evaluates historical persistence; and Glottolog and Dyen change the linguistic basis used for robustness.

Table 7: Linguistic-measure constructions and their roles. Sources: Census of India 2001 language tables (Office of the Registrar General and Census Commissioner of India 2001); *Language Atlas of India 1991* (Office of the Registrar General and Census Commissioner, India 2004); Shastry (2012); Glottolog 5.3 (Hammarström et al. 2026); and Dyen, Kruskal, and Black (Dyen et al. 1992; “Comparative Indo-European Database” 1997).

Manuscript measure	Concept	Construction	Purpose
Speaker-weighted distance from Hindi	Average relative separation from Hindi	Census-2001 mother-tongue speaker	Primary comparative language-cost measure
	among distance-bearing local languages	weights $\times$ reviewed Shastry 0–5 degrees; Hindi/Urdu and native English do not enter the weighted mean	
	Same Shastry concept using only the largest local languages	Speaker-weighted mean among the three largest mapped mother tongues	Comparison with the original analysis

Manuscript measure	Concept	Construction	Purpose
Share speaking languages distant from Hindi	Population mass facing relatively large Hindi distance	Share of all mother-tongue speakers assigned Shastry degree 3–5	Nonlinear alternative
Language-distance composition	Distribution of local speakers across the Shastry scale	All-speaker shares at degrees 1–5; degree 0 is the joint-model reference	Richer composition sensitivity
Historical speaker-weighted distance from Hindi	Earlier analogue of the preferred measure	1991 Language Atlas speaker counts $\times$ the reviewed Shastry basis, subject to coverage and uncertainty thresholds	Persistence and historical validation
Genealogical distance from Hindi	Separation in a modern language-family classification	Census speaker weights $\times$ reviewed Glottolog 5.3 genealogical edge distance	Genealogical robustness
Lexical noncognacy with Hindi	Dissimilarity in basic vocabulary	Census speaker weights $\times$ (100 – Dyen cognate percentage with Hindi)	Lexical robustness

A.4 Price Series and Consumption Harmonization

Comparing district consumption across two decades requires more than placing several survey rounds in one file. The historical NSS rounds and the Household Consumption Expenditure Surveys (HCES) of 2022–23 and 2023–24 differ in questionnaire design, recall periods, district codes, and price environments ([National Sample Survey Office 2007](#), [2012a](#), [2012b](#), [2014](#), [2024](#), [2025a](#); [National Statistical Office 2011](#)). I therefore reconstruct household monthly per-capita consumption expenditure (MPCE) within each survey first, validate that nominal measure against the published survey totals, and only then harmonize prices and geography. The outcome used in the paper is real consumption per person, not a nominal rupee comparison across survey years.

For the NSS Schedule 1.0 rounds, I retain the consumption concept reported for that design rather than forcing every round into a nominally identical recall rule. The source metadata identify the household MPCE field, its scale, and the survey-design variables required for weighting; the benchmark metadata separately record whether the published comparison is based on the mixed reference period (MRP) or modified mixed reference period

(MMRP). Under MRP, less frequently purchased categories use a 365-day recall while most other spending uses 30 days; MMRP additionally uses a 7-day recall for selected food and related items ([National Sample Survey Office 2025b](#)). Some releases report MPCE directly; others require reconstruction from the split household files. I check the resulting household series against the official rural and urban MPCE for every registered round before any price adjustment. Across the nine surveys used here, the 18 reconstructed rural/urban means differ from their published counterparts by less than 0.50 in every case. A mismatch stops construction of that round rather than being carried into the district estimates.

The modern HCES releases require an additional reconstruction because they replace the single long consumption interview used by earlier NSS rounds with three questionnaires administered in separate visits: the Food Questionnaire (FDQ), Consumables and Services Questionnaire (CSQ), and Durable Goods Questionnaire (DGQ) ([National Sample Survey Office 2024, 2025b](#)). The questionnaires use reference periods of 7, 30, or 365 days depending on the item. I identify the registered expenditure items in each questionnaire, convert each reported amount to a monthly equivalent using its stated reference period, and then sum the three questionnaire components. Imputed house and garage rent is explicitly excluded from this MPCE reconstruction, consistent with the official published MPCE concept used for the benchmark comparison. The 2023–24 release identifies visit order, so I use the multiplier from the third visit, as prescribed in the official estimation procedure ([National Sample Survey Office 2025b](#)). The 2022–23 release does not preserve visit order in the same way; its three questionnaire records are therefore required to carry the same multiplier before I use their common value. Households missing one of the three questionnaire components are not treated as complete HCES consumption observations.

Nominal MPCE must then be made comparable across places and dates. I separate this adjustment into a **temporal price relative**, which measures price change through time within a state and rural or urban sector, and a **spatial price relative**, which measures price-level differences across state-sector cells in the reference year. Before 2013, rural temporal prices use the Labour Bureau’s Consumer Price Index for Rural Labourers (CPI-RL), whose 1986–87-base series covers 20 states and all-India ([Labour Bureau, Government of India 2026b](#)). Urban prices use the Labour Bureau’s 2001-base Consumer Price Index for Industrial Workers (CPI-IW), aggregated from its centre series to the state level ([Labour Bureau, Government of India 2026a](#)). From 2013 onward, I use the state rural and urban CPI series with base 2012=100 from the national CPI system ([Reserve Bank of India 2026](#)). I link the pre-2013 series to the 2012-base state CPI using their common state-sector months in 2013–14; state-sector cells without sufficient overlap fail rather than receiving an assumed link. A separate overlap between the 2010- and 2012-base rural/urban CPI series places the July 2011–June 2012 reference-year index on that same scale.

I preserve cross-state and rural/urban price-level differences using the state-specific 2011–12 Tendulkar poverty

lines (Planning Commission, Government of India 2013). Each state-sector poverty line is divided by the all-India rural reference of 816 to obtain the spatial relative; using a common denominator retains the national rural-urban price difference rather than normalizing it away. Direct state observations are preferred throughout. When a historical index is unavailable for a small state or union territory, the price crosswalk permits only documented donor or all-India substitutions; the source and reason remain attached to the resulting household deflator. The household price deflator is therefore

$$P_{srt} = P_{sr}^{\text{spatial}} \times P_{srt}^{\text{temporal}},$$

where s denotes state, r rural or urban sector, and t survey time. Real MPCE is nominal MPCE divided by this deflator. For NSS rounds, I average the monthly state-sector indices over the three months represented by the household’s survey sub-round. For HCES, the price assignment follows the household’s three-month panel. This timing avoids using an annual price index when the fieldwork itself identifies a narrower interview period.

The final step is geographic and statistical. I attach each household’s survey district to the Census-2001 geography using the lineage system described in Section A.1. Unresolved or ineligible source geography is recorded rather than silently assigned. Person-weight coverage ranges from about 92.6% in 2000–01 to complete coverage in 2007–08, is about 97–98% in 2009–12, and is about 91.6% in each modern HCES round. District means are person-weighted: a household’s survey weight is multiplied by household size so that the district mean represents consumption per person rather than the mean of equally weighted households. The district welfare system also records the number of sampled households and primary sampling units and a Kish effective sample size, which declines as survey weights become more unequal, and it applies predeclared support thresholds before an estimate enters the preferred analysis. Comparisons that require the same districts use explicit common-support samples rather than allowing the set of districts to change unnoticed across outcomes or rounds.

The two modern HCES rounds provide a useful internal check because they use the same three-visit design in adjacent years. Figure 6 compares district estimates on common preferred support. The correlation is 0.853 for real mean MPCE, 0.852 for mean log real MPCE, and 0.855 for weighted median real MPCE. These correlations do not make the modern HCES directly interchangeable with the older NSS designs; they show that the reconstruction produces similar district rankings when the survey design is held nearly fixed across adjacent rounds. Together with the national MPCE benchmark checks, this is why the paper treats consumption as a harmonized multi-survey measurement system rather than as one nominal outcome carried unchanged across time.

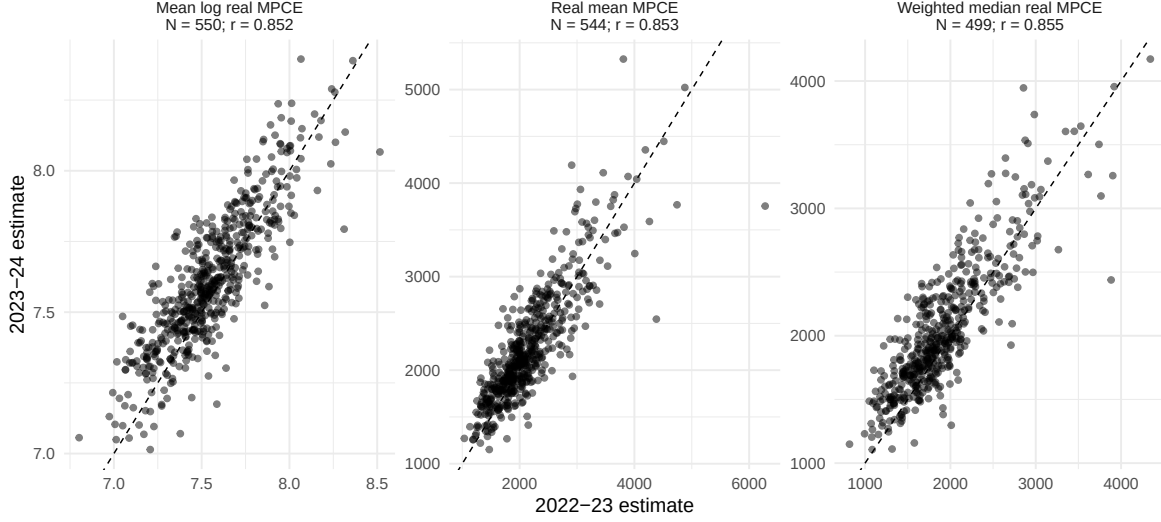


Figure 6: District-level consistency of reconstructed consumption across the 2022–23 and 2023–24 HCES rounds. Each panel uses districts satisfying the preferred support rule in both rounds; the dashed line is the 45-degree line. Sources: National Sample Survey Office, HCES 2022–23 and HCES 2023–24 ([National Sample Survey Office 2024](#), [2025a](#)).

B Migration

B.1 Literature Review

Migration changes the interpretation of district outcomes. The schooling measures in this paper describe children living in a district in 2007–08, while later consumption describes the population living there many years afterward. If people relocate, the economic trajectory of a place need not equal the later economic trajectory of the children originally observed there. I therefore interpret the long-run district outcomes as place-based outcomes for changing populations, not as individual returns to the children exposed to a particular schooling environment.

Earlier work suggested that this distinction might be quantitatively modest in India. Chakraborty and Bakshi (2016) report low interdistrict mobility in West Bengal, Haryana, and Punjab and approximate an individual’s district of residence or employment with the district of schooling. Using earlier NSS rounds, Topalova (2005) similarly documents low decadal interdistrict mobility, especially in rural areas and for employment-related moves. These measures describe recent-duration migration rather than the accumulated stock of everyone who has ever migrated; the distinction matters because lifetime-migrant measures accumulate moves over a longer exposure window as respondents age and can be dominated by moves made long before the economic period under study (Mahapatro 2020).

I do not assume that mobility remained negligible. Evidence using Census cohorts and railway passenger flows

points to substantially more interdistrict movement after 2001 than the late-twentieth-century evidence implied (Economic Divison 2015). More recent work also shows why movement can respond to skill-intensive opportunity. Ghose (2026) studies India’s information-technology expansion in a setting where education and internal migration respond endogenously, so regional welfare gains depend in part on who acquires skills and where workers relocate. At the same time, migration remains structured by institutions. Caste-based rural insurance networks can discourage migration when leaving weakens access to informal insurance (Munshi and Rosenzweig 2016), while state borders themselves sharply reduce migration between otherwise neighboring districts (Kone et al. 2018). These findings make migration neither negligible nor frictionless.

My migration measures come from the 2011 Census D-series tables (Office of the Registrar General and Census Commissioner, India 2011). D-02 describes migrants by place of last residence and provides the interstate share among migrants. D-03 records reason for migration. For work-related sorting I use D-07, which identifies migrants who have lived at the destination for at most nine years, report work or employment as the reason for migration, and previously lived elsewhere in India. I refer to this group as **recent work migrants**. The D-07 total is required to agree exactly with the corresponding D-03 work-migrant subtotal before the measures are used. Counts from the 2011 district geography are then pooled to the Census-2001 geography only where the district lineage is complete and deterministic, as described in Section A.1.

The distinction between all migrants and recent work migrants is important for the paper’s economic question. Overall migrant composition combines moves for marriage, education, household relocation, and work. Restricting attention to recent work migrants instead asks whether destinations differ in the kinds of workers they recently attracted for employment. The result therefore concerns the composition of in-migration observed at the destination; it does not measure the skills of people who left the district.

B.2 Empirical Results

Table 8 summarizes the preferred within-state reduced forms. Each coefficient compares districts within states and conditions on the predetermined Census controls used elsewhere in the paper. The first panel retains broad migration margins even when their estimates are null. The second focuses on recent work migrants and includes both their geographic origin and skill composition, together with the predeclared Hindi-belt restriction.

The broad mobility measures provide little evidence that linguistic distance is associated with simply receiving more geographically distant migrants. The separate 2001–11 changes in population-denominated migrant rates remain descriptive and are not part of this predeclared regression family, so I do not use them to make an inferential claim about total migration incidence here. The interstate share among all migrants is close to zero in the preferred specification. The work- and education-reason shares provide no family-adjusted evidence of an

Table 8

Linguistic Distance and Migration Composition. Source: Census of India 2011 D-02, D-03, and D-07 migration tables.

	(1)	(2)	(3)
<i>Panel A: All migrants</i>			
Linguistic distance from Hindi	-0.0023	0.0060	-4.329e-04
	(0.0166)	(0.0050)	(0.0010)
Dependent variable	Interstate share among migrants	Work/employment share among migrants	Education share among migrants
Sample	All states	All states	All states
Fixed effects	State	State	State
Predetermined controls	Yes	Yes	Yes
Observations	355	355	355
<i>Panel B: Recent work migrants</i>			
Linguistic distance from Hindi	0.0050	0.0129**	0.0079
	(0.0266)	(0.0044)	(0.0057)
Dependent variable	Outside-state share among recent work migrants	Graduate/technical-degree share among recent work migrants	Graduate/technical-degree share among recent work migrants
Sample	All states	All states	Hindi-belt states
Fixed effects	State	State	State
Predetermined controls	Yes	Yes	Yes
Observations	355	355	178

Each column is a separate district regression using speaker-weighted linguistic distance from Hindi, predetermined Census-2001 controls, state fixed effects, and state-clustered standard errors. National significance stars use Holm-adjusted p-values across the eight registered migration outcomes in the preferred specification; the predeclared Hindi-belt restriction uses its single raw p-value. * p < 0.10, ** p < 0.05, *** p < 0.01 The dependent variables are shares, so coefficients are changes in shares associated with a one-degree increase in linguistic distance. These are conditional associations, not causal migration effects.

association. Among recent work migrants, the share arriving from outside the state is also near zero. These null results are useful because they distinguish a change in migrant composition from a general increase in longer-distance mobility.

The skilled composition result is different. A one-degree increase in speaker-weighted linguistic distance from Hindi is associated with a 0.0129 increase in the graduate-or-technical-degree share among recent work migrants; the state-clustered standard error is 0.0044 and the Holm-adjusted p-value is 0.027. I interpret this as evidence of skill sorting across destinations, not as evidence that linguistic distance causes migration or that migrants mediate an EMI effect. The dependent variable is the skill composition of recent work in-migrants conditional on being observed at the destination.

The predeclared Hindi-belt restriction is weaker. Within the Hindi-belt states, the corresponding estimate is 0.0079 with a standard error of 0.0057 ($p = 0.171$). The national within-state association therefore is not clearly reproduced inside that more linguistically similar institutional setting. This boundary is one reason I do not reduce the migration evidence to a universal behavioral response to linguistic distance.

Taken together, the migration results sharpen the interpretation of the later district outcomes. Linguistic conditions are associated with **who arrives for work** more clearly than with broad migration composition or interstate movement. Migration can therefore change the skill composition of places even when aggregate mobility margins show little association. Later district consumption should consequently be read as local development for a changing population, not as the individual economic return to the children observed in the earlier schooling data.

C Education Selection

The district comparisons in the main text show that schooling access differs sharply by social group even within the same local market. I retain the original child-level enrollment model here as a complementary description of *who is enrolled at all*. The model does not identify a causal effect of any household characteristic or schooling program, and I do not use it as a control function or selection correction in the district analysis.

I use the NSS 64th Round *Participation and Expenditure in Education* survey, restricting the child roster to ages 5–19 ([National Sample Survey Office 2011](#)). The outcome equals one when the child appears in the survey’s enrolled-child education record and zero otherwise. The fitted probit includes age, sex, household size, religion, social group, rural or urban residence, distance to the nearest primary class, and the education of the child’s father or the registered male household proxy. It also includes district averages of selected schooling-cost and benefit variables among enrolled children. Those district averages are useful adjustment variables for reproducing the original model, but they are partly determined by school provision and the composition of enrolled children.

I therefore keep them in the fitted specification while omitting them from the reader-facing coefficient table.

The final specification uses the NSS survey design: household weights, first-stage sampling units corresponding to sampled villages or urban blocks, and the survey’s strata and substrata. Because a probit coefficient is expressed on a latent index, I report quantities on the probability scale. For continuous variables, the table reports the average slope of the predicted enrollment probability. For categorical variables, it reports the average discrete change in predicted enrollment probability when each observed child is assigned the comparison category rather than the stated reference category, with the remaining covariates left at their observed values. These are average marginal quantities over the estimation sample rather than effects evaluated at a synthetic “average child” (Hansen 2022; Arel-Bundock et al. 2024).

Table 9: Correlates of School Enrollment: Average Marginal Effects from a Survey-Weighted Probit. Source: NSS 64th Round, 2007-08, Participation and Expenditure in Education.

	Enrolled (1 = yes)
	(1)
Age (years)	-0.028*** (0.000)
Female (ref: Male)	0.044*** (0.003)
Household size	-0.004*** (0.001)
Religion: Muslim (ref: Hindu)	-0.126*** (0.007)
Religion: Christian	-0.043*** (0.014)
Religion: Sikh	0.040 (0.025)
Religion: Jain	-0.001 (0.012)
Religion: Buddhist	0.047* (0.024)
Religion: Zoroastrian	0.101*** (0.034)
Religion: Other	0.252*** (0.003)
Social group: Scheduled Tribe (ref: Other)	-0.102*** (0.008)
Social group: Scheduled Caste	-0.075*** (0.006)
Social group: Other Backward Class	-0.029*** (0.005)
Urban (ref: Rural)	-0.003 (0.005)
Distance 1–2km (ref: <1km)	-0.077*** (0.026)

Table 9: Correlates of School Enrollment: Average Marginal Effects from a Survey-Weighted Probit. Source: NSS 64th Round, 2007-08, Participation and Expenditure in Education. *(continued)*

	(1)
Distance 2–3km	-0.001 (0.010)
Distance 3–5km	0.004 (0.066)
Distance > 5km	0.036 (0.035)
Father’s educ.: Literate, no school (ref: Illiterate)	0.097*** (0.017)
Father’s educ.: Literate, school < primary	0.127*** (0.006)
Father’s educ.: Primary	0.142*** (0.005)
Father’s educ.: Upper primary	0.195*** (0.006)
Father’s educ.: Secondary	0.212*** (0.007)
Father’s educ.: Higher secondary	0.247*** (0.008)
Father’s educ.: Postsecondary+	0.245*** (0.009)
Observations	114898

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ Entries are average changes in predicted enrollment probability. Continuous covariates are reported as average slopes; categorical covariates are average discrete comparisons against the stated reference category, evaluated over the observed covariate distribution. The probit uses NSS survey weights and design-based standard errors clustered at the primary sampling unit. District-level schooling-context controls constructed from enrolled children remain in the fitted model but are omitted from the table and are not interpreted causally.

The social-identity results reinforce the main-text evidence that access to education is not evenly distributed. Relative to Hindu children with the same modeled covariates, Muslim children have an average predicted enrollment probability about 12.6 percentage points lower; the corresponding Christian comparison is about 4.3 points lower. The Muslim difference is consistent with evidence of persistent educational disadvantage among Indian Muslims (Borooah and Sabharwal 2021). Relative to children in the residual “Other” social-group category, the predicted enrollment probability is about 10.2 points lower for Scheduled Tribe children, 7.5 points lower for Scheduled Caste children, and 2.9 points lower for Other Backward Class children. I interpret these as conditional associations, not as effects of religion or caste. That caution is especially important in a setting where social institutions and caste networks can alter schooling choices and the occupational opportunities attached to them (Munshi and Rosenzweig 2006).

Parental education displays the largest systematic gradient. Relative to an illiterate father or male household proxy, the average comparison rises from roughly 9.7 percentage points for a literate adult with no formal schooling to about 14.2 points for primary education, 21.2 points for secondary education, and 24.5 points for postsecondary education. Father’s education can proxy for several household characteristics at once, including permanent income, information about schooling and labor markets, and preferences or expectations transmitted across generations. The model cannot separate those channels. What it establishes is that enrollment is strongly stratified by intergenerational human capital even after conditioning on the other observed covariates.

Several other estimates are useful mainly as descriptive boundaries. Enrollment probability declines by about 2.8 percentage points per year of age across the 5–19 range. Girls have a roughly 4.4-point higher predicted probability than boys conditional on the other modeled characteristics; I do not attribute that difference to any particular program, although Sarva Shiksha Abhiyan and girls-focused schooling initiatives were already operating before the survey ([Press Information Bureau 2008](#)). Each additional household member is associated with a small decline in predicted enrollment probability. That sign is consistent with the resource-dilution logic of quantity–quality models, but Indian evidence on whether family size causally reduces child investment is mixed ([Becker 1969](#); [Kumar and Kugler 2011](#); [Onur and Velamuri 2021](#)).

School access is also unevenly related to enrollment. Children whose nearest primary class is reported one to two kilometers away have a predicted enrollment probability about 7.7 percentage points lower than children with a primary class within one kilometer. The more distant categories are estimated much less precisely and do not form a monotone distance gradient. I therefore treat the one-to-two-kilometer comparison as evidence that local access conditions can matter, not as an estimate of a stable distance-response function.

The original model also includes district averages of free schooling, fee waivers, scholarships, textbooks, stationery, mid-day meals, and enrollment cost. I do not display or interpret those coefficients here. These variables are measured among enrolled children and are jointly determined by program targeting, school type, local provision, and household selection. In particular, the negative coefficient on the district share receiving free education cannot establish that free schooling reduces enrollment. Retaining those variables as controls preserves the original specification without turning selected program receipt into a policy effect.

The child roster contains 127,181 observations, but the probit is estimated on complete cases. Missingness is concentrated in three required covariates: father’s education, district-average enrollment cost, and distance to the nearest primary class. The registered missingness checks show that approximately 9.7 percent of children are missing at least one required model covariate. Missingness is also predictable from observed characteristics, especially for district-average enrollment cost and school distance, so the estimation sample should not be treated as a random subsample of all children.

Table 10

Missingness in the Child-Enrollment Probit. Source: NSS 64th Round, 2007-08, Participation and Expenditure in Education.

Variable	Missing N	Missing %	Missingness pseudo-R2
Distance to nearest primary class	312	0.2	0.246
District-average enrollment cost	7,109	5.6	0.310
Father's education	5,203	4.1	0.023
Any model covariate	12,283	9.7	

Note: The final row reports children missing at least one covariate required by the probit. Pseudo-R2 values come from the registered missingness-logit screen; the blank aggregate cell has no corresponding missingness regression. No missing-value imputation is used.

This limitation is another reason I use the model only as corroborating descriptive evidence. Its central message agrees with the within-district schooling-access analysis in the main text: educational participation is strongly associated with social identity and family background. The district decomposition is more directly suited to the paper's question about access to English-oriented schooling; the probit shows that unequal selection begins one step earlier, at entry into schooling itself.

D Linguistic Distance as an Instrumental Variable

The main text uses linguistic distance principally as a measure of inherited linguistic conditions. I separately ask whether it can serve as an instrumental variable for English-medium schooling: that is, whether variation in linguistic distance can isolate variation in English-medium exposure that is sufficiently strong and otherwise excluded from the later consumption equation. The answer is no. The purpose of this appendix is to show where the design fails and why changing the linguistic measure does not repair it. This distinction is important because weak instruments can make conventional two-stage least squares (2SLS) estimates and their standard errors unreliable even when a familiar first-stage threshold appears to be met (Keane and Neal 2024).

D.1 First Stage Collapses

The broad national relationship between linguistic distance and English-medium schooling is not itself in doubt. With the preferred speaker-weighted distance and the predetermined Census controls, but without geographic fixed effects, the excluded-instrument F statistic is 14.29. Adding region fixed effects reduces it to 3.24; adding state fixed effects instead reduces it to 0.75. The same attenuation is summarized alongside the alternative linguistic measures in Table 11. The corresponding state-fixed-effects partial R^2 is 0.0039: after comparing districts within the same state and conditioning on predetermined characteristics, linguistic distance explains very little of the remaining variation in all-child English-medium exposure.

The distinction between the conventional first-stage F statistic and the Montiel Olea–Pflueger effective F statistic matters here. The conventional statistic is useful for comparing the same first-stage specification as geographic controls are added. The effective F is designed for assessing weak-instrument relevance when errors may be heteroskedastic or clustered (Montiel Olea and Pflueger 2013). I therefore use the conventional F statistics to describe geographic absorption and the effective F statistic when evaluating the preferred IV specification. I do not treat a value of 10 as a universal pass/fail rule; contemporary weak-IV guidance shows that important distortions in 2SLS inference can persist well above that threshold (Keane and Neal 2024).

The first-stage collapse is substantively informative. Linguistic distance and English-medium schooling are strongly related across broad Indian geography, but most of that relationship is absorbed when districts are compared within states. This fits the institutional evidence in the main text: English-medium schooling, especially in government schools, is strongly organized at the state level. It also means that linguistic distance cannot provide the within-state excluded variation needed to interpret the later schooling–consumption association as a causal effect of English-medium instruction.

Table 11

First-Stage Relevance Across Linguistic Measures. Sources: Census of India 1991 and 2001; NSS 64th Round, 2007-08.

Linguistic measure	No geographic FE F	Region FE F	State FE F	State partial R ²	N
Speaker-weighted distance from Hindi	11.73	3.24	0.75	0.004	573
Top-three-language distance from Hindi	16.30	4.65	1.96	0.012	573
Genealogical distance from Hindi	10.43	0.32	0.28	0.001	573
Lexical noncognacy with Hindi	5.29	0.07	0.15	0.000	573
Share speaking languages distant from Hindi	4.08	0.54	0.59	0.004	573
Language-distance composition (5 instruments)	102.28	22.37	3.73	0.041	573
Historical 1991 speaker-weighted distance	9.98	0.31	0.03	0.000	89

Note: The first three columns report the state-clustered joint excluded-instrument F statistic under no geographic fixed effects, six-region fixed effects plus the main predetermined controls, and state fixed effects plus the same controls. The final relevance column is the state-FE partial R-squared. The historical row uses its validated 1991 common support and is therefore not numerically pooled with the modern district sample. These conventional first-stage statistics describe relevance only; they are not substituted for the Montiel Olea-Pflueger effective-F statistic used for weak-IV assessment.

D.2 Robustness to Measures and Data

The weak within-state relationship is not specific to one way of measuring linguistic distance. Table 11 compares the preferred speaker-weighted measure with the legacy top-three measure, genealogical distance, lexical noncognacy, the population share speaking languages distant from Hindi, a five-component language-distance composition, and the independently reconstructed 1991 distance measure. For each construction, I report the excluded-instrument F statistic without geographic fixed effects, with region fixed effects, and with state fixed effects. The last column reports the state-fixed-effects partial R^2 .

The scalar alternatives all tell the same story. The top-three measure has a raw F of 16.30 but a state-fixed-effects F of 1.96. The genealogical and lexical measures fall from 10.43 and 5.29 to 0.28 and 0.15, respectively, while the distant-language share falls from 4.08 to 0.59. The richer five-component language composition is strongly jointly predictive without geographic fixed effects, but its state-fixed-effects joint F is only 3.73. A richer set of language shares can therefore predict broad geographic differences without supplying strong within-state excluded variation.

Moving the linguistic measure backward in time does not solve the problem either. On the validated common historical support, the 1991 speaker-weighted distance has an F statistic of about 9.98 without geographic fixed effects, 0.31 with region fixed effects and Census controls, and 0.03 with state fixed effects and those controls. The historical construction is valuable because it separates measurement of the linguistic environment from later

schooling, but it does not restore relevance. The broader historical checks reach the same boundary: linguistic geography is persistent, yet predeterminedness and persistence do not imply that the exclusion restriction holds or that the first stage is strong. Pretreatment balance and negative-control checks can inform an IV design, but they cannot by themselves establish exclusion, and their interpretation also depends on specification choices (Danieli et al. 2026).

Measurement error in English-medium schooling is also an implausible explanation for the collapse. The independently reconstructed DISE measure agrees closely with the NSS measure in levels and continues to agree after state means are removed; Figure 5 reports correlations of about 0.896 and 0.607, respectively. The language-measure sensitivities likewise permit alternative crosswalks and language classifications, yet none produces a persuasive state-level first stage. These results make the identification boundary robust to the two most immediate measurement objections: changing the schooling source and changing the linguistic-distance construction.

The richer language-share instruments do not offer a credible escape through multiple instruments. In the registered state-fixed-effects specifications, constituent first stages remain weak and the falsification-adaptive sets contain zero. A falsification-adaptive set summarizes parameter values that remain compatible with the model under minimally sufficient relaxations of overidentifying restrictions (Masten and Poirier 2021). Here it reinforces the simpler relevance evidence rather than recovering a stable causal estimate.

D.3 Weak-IV-Robust Inference

Weak first stages change how the second stage should be read. Conventional 2SLS estimates remain useful as a description of what the ratio estimator produces, but their usual t tests are not the basis for inference. I instead emphasize the Montiel Olea–Pflueger effective F statistic and Anderson–Rubin (AR) tests. The AR test evaluates candidate values of the structural coefficient without requiring a strong first stage and is therefore appropriate when identification is weak (Keane and Neal 2024).

Table 12 reports the two long-change specifications used in the main identification discussion. I do not attach conventional significance stars to the 2SLS coefficients. The table instead places the weak-IV statistics directly beside them.

For 2004–05 to 2022–23, the 2SLS coefficient is 0.117 with a state-clustered standard error of 0.152; the effective F statistic is 0.69. For the 2023–24 endpoint, the corresponding estimate is 0.137 with a standard error of 0.234 and an effective F of 0.41. Neither conventional estimate is informative enough to support a causal interpretation. More importantly, the AR confidence sets are disconnected. On the evaluated grid, the 2022 set accepts values at or below approximately -0.071 and at or above 0.025 , while the 2023 set accepts values at or below approximately -0.051 and at or above 0.015 . Both sets extend to the grid boundaries. Rejecting zero on such a grid does not

Table 12

Weak-IV-Robust Long-Run Inference. Sources: Census of India 2001; NSS 61st Round, 2004-05; NSS 64th Round, 2007-08; HCES 2022-23 and 2023-24.

	2022-23 long change	2023-24 long change
EMI exposure	0.117 (0.152)	0.137 (0.234)
MOP effective F	0.69	0.41
AR p-value at zero	0.001	0.014
AR 95% confidence set	[grid ≤ -0.193146, -0.0714641] U [0.025109, 0.193146 ≤ grid]	[grid ≤ -0.185982, -0.0511449] U [0.0148785, 0.185982 ≤ grid]
Direct effect needed to admit zero	41.7%	19.9%
Observations	524	522

Columns report conventional 2SLS coefficients with state-clustered standard errors in parentheses. No significance stars are shown because weak-IV-robust inference takes precedence over conventional 2SLS t tests. MOP effective F is the Montiel Olea-Pflueger weak-instrument statistic. AR denotes Anderson-Rubin inference; the reported 95

identify the sign of the effect: both negative and positive values remain compatible with the weak-IV-robust test.

The result is not a peculiarity of these two specifications. Across every registered consumption-IV robustness family, no specification satisfies the project’s predeclared effective-F strength criterion. The families vary the linguistic measure, English-medium treatment, welfare outcome, controls, parameterization, and historical adjustment. I retain the full model-level results in the replication outputs rather than printing the hundreds of specifications here. Their collective implication is narrower and more useful: there is no nearby specification choice that turns the design into a strongly identified IV strategy.

Finally, I relax exact exclusion rather than treating it as guaranteed. Following the logic of plausibly exogenous instruments, I allow linguistic distance to have a bounded direct effect on the outcome and ask how large that direct effect must be before $\beta = 0$ enters the AR confidence set (Conley et al. 2012). For the 2022 long change, the minimum bound is about 0.0267 log points, or 41.7% of the observed reduced form. For 2023, the corresponding threshold is about 0.0112 log points, or 19.9% of the reduced form. These are sensitivity thresholds, not estimates of a direct effect. They show that the 2023 result is especially fragile to modest departures from exact exclusion, while even the more persistent 2022 reduced form does not overcome the weak first stage.

I therefore do not use linguistic distance to estimate a causal effect of English-medium schooling. The appendix supports a more limited conclusion: linguistic distance is a persistent and economically meaningful feature of India’s language environment, but it is too weakly related to within-state English-medium schooling, and too plausibly related to broader local conditions, to isolate that schooling channel. The descriptive schooling–welfare relationships in the main paper remain descriptive.

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